

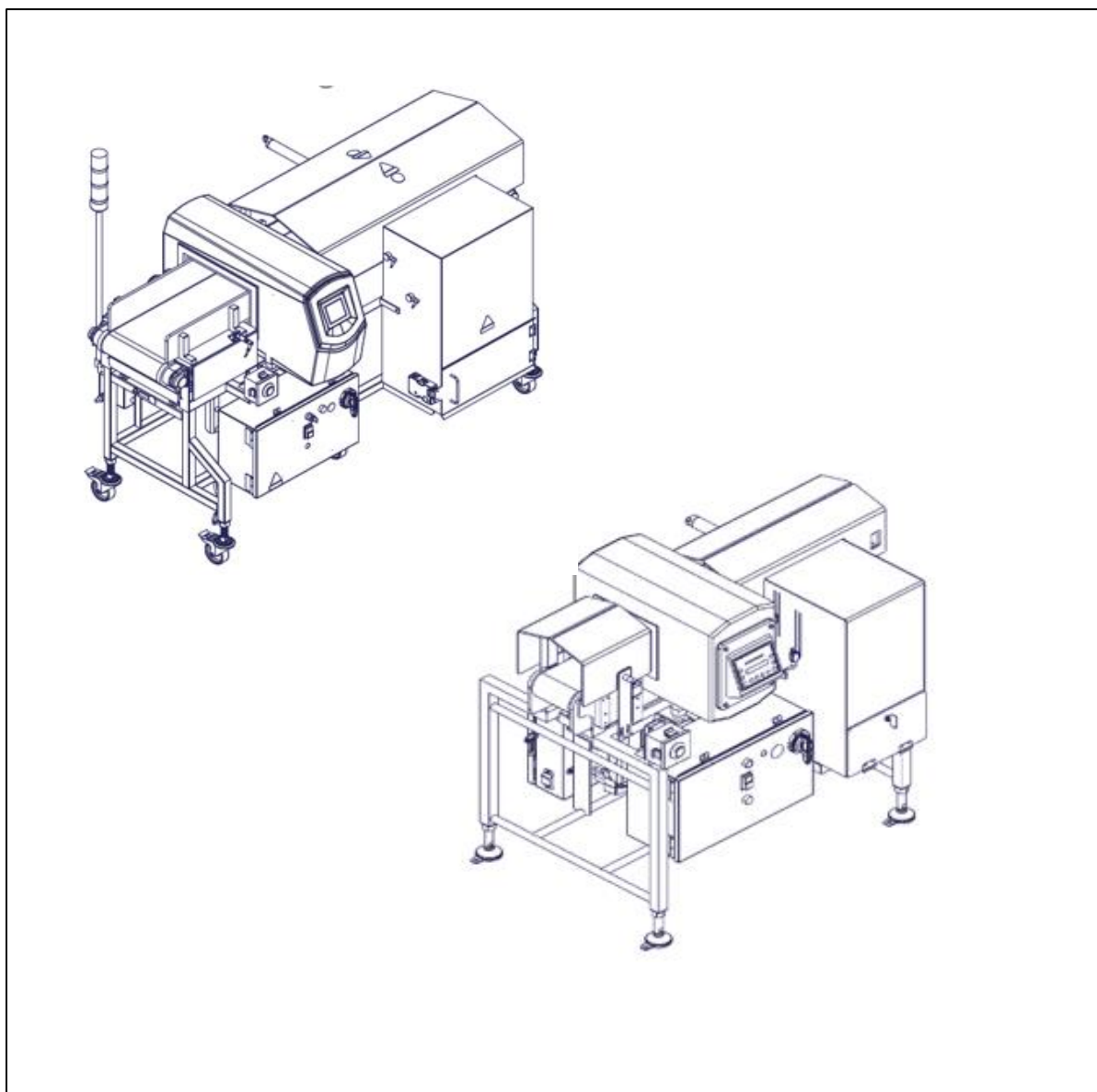


Table of Contents

1	Introduction	3
1.1	Intended Audience	3
1.2	To Protect Your Metal Detection System's Future	4
1.3	How To Use This Manual	4
2	Safety Notes	4
2.1	Safety Labels	5
2.2.1	Hazard Notifications	5
2.2.1.1	Definitions of Signal Words	5
2.2.1.2	Meaning of Hazard Alert Symbols	6
2.2.2	Mandatory Procedures	6
2.2.3	Prohibited Procedures	6
2.3	General Protective Procedures	7
2.4	Safety Notes for Various Activities	7
2.4.1	Transporting and Setting the Equipment in Place	7
2.4.2	Installing	10
2.4.3	Operating the Equipment and Monitoring the Inspection Process	10
2.4.4	Testing and Verifying the Equipment	10
2.4.5	Maintaining, Cleaning and Sanitizing the Equipment	10
2.5	Special Hazards	10
2.5.1	Electricity	10
2.5.2	Pneumatics	11
2.5.3	Moving Parts and Hot Parts	11
2.5.4	Rejectors	11
2.5.5	Noise	11
2.5.6	Electromagnetic Fields	12
2.5.7	Lights and Laser Sensors	12
2.5.8	Strobe Lights	12
2.5.9	X-ray Radiation	12
2.5.10	Explosive Atmospheres or Locations	12
3	Design and Function	13
3.1	Options for Metal Detection Systems	13
3.2	Components of a Stop-on-Detect System	15
3.3	How the Stop-on-Detect System Works	17
3.4	Additional Components of an Automatic Reject System	18
3.5	How the Automatic Reject System Works	21
3.6	Additional Components of a Due Diligence System	22
3.7	How the Due Diligence System Works	24
4	Installation	26
4.1	Transporting and Unpacking the System	26
4.2	Lifting and Moving the System	27
4.2.1	Lifting the System Using a Forklift Truck	27
4.2.2	Lifting the System Using Powered Lifting Equipment	27
4.3	Positioning the System	27
4.4	Adjusting the Height of the System	28
4.4.1	Leveling the Conveyor	29
4.4.2	Adjusting the Feet	30
4.4.3	Locking and Unlocking the Casters	31
4.4.4	Adjusting the Height of the Lockable Casters	31
4.5	Metal-Free Zone (MFZ)	32
4.6	Avoiding Mechanical Vibration	32
4.7	Connecting the Electrical Power Supply	32
4.8	Connecting the Compressed Air Supply	33
4.9	Commissioning the System	35
5	Operation	35
5.1	Before You Start the System	35
5.1.1	Inspecting Flat PU and Modular Conveyor Belts and Rollers	35
5.1.2	Checking the Guarding and Fittings	36
5.1.3	Checking the Reject Bin	36
5.1.4	Examining the Safety Labels	36
5.1.5	Checking the Electrical Power Supply Cabling	36

5.1.6	Checking Sensors, Light Beams and Cabling	36
5.1.7	Inspecting the Liner of the Metal Detector's Aperture	37
5.1.8	Checking the Motor with Gearbox Unit	37
5.1.9	Inspecting the Pneumatic System	37
5.2	Starting, Stopping and Isolating the System	38
5.2.1	Starting and Stopping a Stop-on-Detect System	38
5.2.2	Starting and Stopping an Automatic Reject System	38
5.2.3	Starting and Stopping a Due Diligence System	39
5.2.4	Using the Emergency Stop Button	40
5.2.5	Isolating the System's Electrical Power Supply	40
5.2.6	Isolating the Compressed Air Supply to the Reject Mechanism	41
5.3	Understanding Warning Systems and Devices	42
5.3.1	Warning Systems for Stop-on-Detect Systems	42
5.3.2	Warning Systems for Automatic Reject and Due Diligence Systems	42
5.4	Hand-Feeding Products onto the Conveyor	43
6 Routine Maintenance and Testing		43
6.1	Testing the System During Production	44
6.1.1	Checking the Belt Tracking	44
6.1.2	Testing Automatic Reject Mechanisms	44
6.1.3	Testing the Reject Confirmation Unit (RCU) sensor	44
6.2	Routine Performance Monitoring	45
6.3	Cleaning and Caring for the System	46
6.3.1	Stopping the System and Doing Lockout and Tagout	47
6.3.2	Using Detergents	47
6.3.3	Cleaning Procedure	48
6.3.3.1	Recommended Precautions	48
6.3.3.2	Wash-down Procedure	48
6.3.4	Restarting the System after Sanitization	49
6.4	Storing the System	49
6.5	Disposing of the System	50
7 Troubleshooting		50
7.1	Conveyor Does Not Run	50
7.2	Testing the Conveyor Belt for Metal Contamination	51
8 Technical Data		51

Disclaimer

In no event shall METTLER TOLEDO be liable for any Damages or injury of any nature or kind, regardless of cause, that arise from the use of the equipment referred to in this manual, except to the extent explicitly agreed otherwise by METTLER TOLEDO in a written document, legally binding upon it.

Strict compliance with the safety procedures established and referred to in this manual and extreme care in the use of the equipment are essential to avoid or minimize the chance of personal injury or damage to equipment.

The information, figures, illustrations, tables, specifications, and schematics contained in this manual are believed to be correct and accurate at the date of publication or revision. However, no representation of warranty with respect to such correctness or accuracy is given or implied, and METTLER TOLEDO will not, under any circumstances, be liable to any person or corporation for any loss or damages incurred in connection with the use of this manual.

The information, figures, illustrations, tables, specifications, and schematics contained in this manual are subject to change without notice.

Any modification to the equipment or any part thereof, without prior approval of the manufacturer is prohibited.

Any software used with this equipment should be used only for its intended purposes. The user shall not undertake changes, modifications, conversions, translations into another computer language, or copies (except for a necessary backup copy).

METTLER TOLEDO's warranty and liability for any equipment malfunction or damages resulting from any violation of the above-stated prohibitions or caused by non-compliance with the applicable manual and product documentation are expressly excluded.

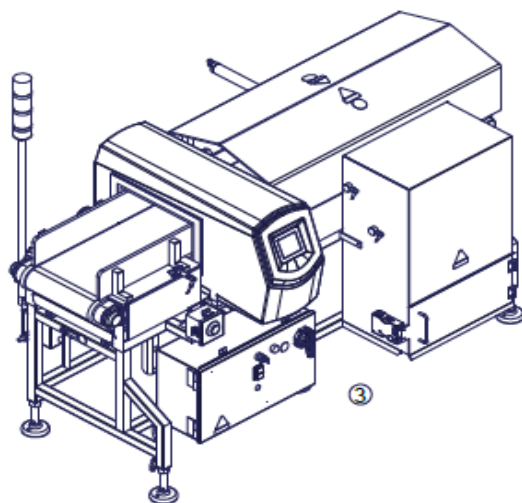
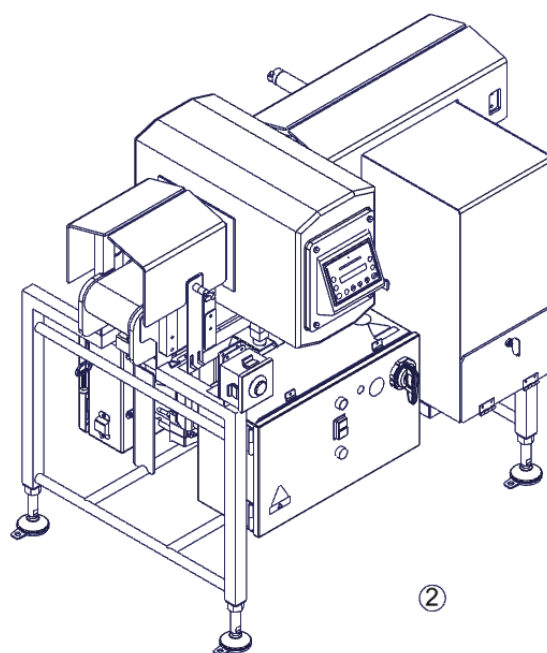
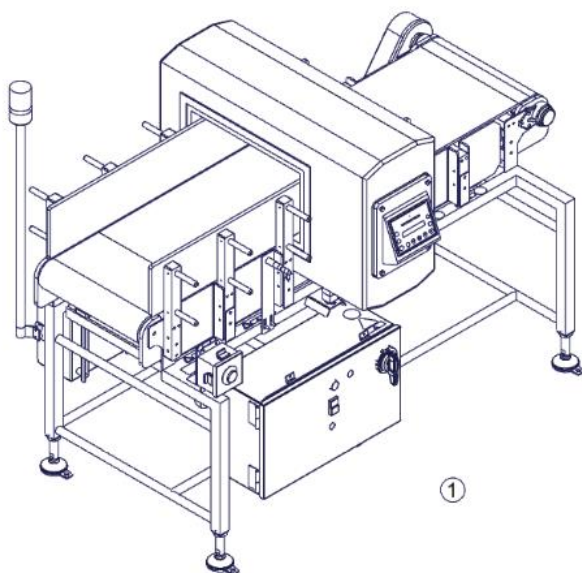
"Damages" includes, but is not limited to all losses and all direct, indirect, incidental, special, consequential and punitive damages arising under a contract, tort or other theory of liability (including reasonable legal and accounting fees and expenses).

1 Introduction

Thank you for buying a METTLER TOLEDO metal detection system. This manual is for the following metal detection systems:

- SL1500/SL2000 stop-on-detect system (1)
- SL1500/SL2000 automatic reject system (2)
- SL1500/SL2000 due diligence systems (3)

The figures below represent typical arrangements of the above metal detection systems:



1.1 Intended Audience

This user manual is for all personnel who transport, install, set up, operate and perform routine maintenance of the metal detection systems we describe.

1.2 To Protect Your Metal Detection System's Future

Contact your authorized METTLER TOLEDO representative about the following products and services:

- Installation and commissioning of your metal detection system
- Certified, genuine parts
- Emergency repair and annual service contracts
- An annual performance verification service to certify that your metal detection system has maintained its performance levels

For more information contact your local service representative: www.mt.com/slmd-contact.

1.3 How To Use This Manual

This manual is organized into the following chapters so that you can find the information you need quickly:

Chapter 1: Introduction

- Tells you what systems are included in the manual.
- Explains who the manual is for and gives a summary of the contents of each chapter.

Chapter 2: Safety Information

- Gives important information about all of the topics that are related to safety.
- Explains how to avoid causing personal injury or damaging the system while you install, operate and perform routine maintenance on the system.

Chapter 3: Design and Function

- Gives illustrations of the main components of the metal detection systems and descriptions of how each system works.

Chapter 4: Installation

- Gives information for materials handling personnel on how to unpack, lift and move the systems.
- Gives information for installation technicians about positioning and leveling the system as well as connecting the compressed air supply and the electrical power supply.

Chapter 5: Operation

- Describes all of the tasks you need to do before you operate the system and how to start and stop the system.
- Explains how to isolate the electrical power and the compressed air supply to the system.
- Helps you to understand how warning systems and devices work.
- Explains how to feed products manually onto the conveyor belt.

Chapter 6: Routine Maintenance and Testing

- Explains how to test the system during production and how to do routine performance monitoring.
- Describes how to care for and clean the system and how to store and dispose of the system.

Chapter 7: Troubleshooting

- A short guide so you can sort out minor faults with the system.

Chapter 8: Technical Data

- Essential technical information about the systems.

2 Important Safety Information



Important Safety Note

Read and understand all the safety notes in the following sections as well as the safety messages in the rest of this manual.

If you do not follow the safety notes and messages, this may lead to property damage and personal injury up to and including death.

2.1 Safety Labels and Notice Labels

The ISO 3864 safety labels are installed at potentially hazardous areas on the equipment. They give special safety-related notifications. The locations of these labels are given in the drawings supplied with your equipment. There are three types of safety labels:

- Hazard notifications
- Mandatory procedures
- Prohibitive procedures

The meanings of the different kinds of safety labels are explained in the following sections.

Before transporting, installing, operating or working on the equipment, find out about the location and meanings of the safety labels. Maintain the safety labels so that they are clear of obstructions and are readable. Do not remove any safety labels. Replace any label that is no longer readable.

2.1.1 Hazard Notifications

A hazard notification consists of the following:

- Hazard alert symbol (yellow triangle with black symbol)
- Signal word (**DANGER**, **WARNING** or **CAUTION**)
- Special notifications related to the hazard (as required)

The signal labels are attached next to the hazard alert symbol labels on the equipment.

2.1.1.1 Definitions of Signal Labels

Signal labels describe the level of risk of a particular hazard. The color of the safety label background indicates the risk, as shown in the following table.

	(red): This hazard severity panel indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
	
	(orange): This hazard severity panel indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
	
	(yellow): This hazard severity panel indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.
	

2.2.1.2 Meaning of Hazard Alert Symbols

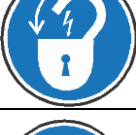
The following hazard alert symbols may be installed on your equipment.

	Crushing
	Shearing
	Cutting
	Entanglement
	Drawing-in, trapping
	Impact
	Ejection of parts
	Abrasion, friction
	High pressure, fluid ejection
	Loss of stability
	Slipping, tripping
	Electrical shock

	Electrostatic
	Non-Ionizing radiation
	Permanent magnetic fields
	Burning, scalding by direct contact, flames or thermal radiation
	Hearing impairment
	Arc flash, explosion
	Laser
	UV light
	Optical radiation
	Ionizing radiation

2.2.2 Mandatory Procedures

Mandatory procedures alert personnel to special required actions. The labels are round with a blue background and a white symbol describing the required action. The following table lists the mandatory procedures labels which may be installed on your equipment.

	Read the user manual
	Read the technical manual
	Wear ear protection
	Lockout, tagout
	Lift point

2.2.3 Prohibited Procedures

Prohibited procedures alert personnel to particular actions to avoid. The labels are round with a circular red band and single diagonal cross bar. The black symbol describes the action to avoid.

The following table lists the prohibited procedure labels that may be installed on your equipment.

	Do not operate without safeguards
	Do not operate the equipment if you have implanted medical devices
	Do not step on the equipment

2.2.4 Notice

The text that follows NOTICE in this manual does not give safety information. However, it is still important because it informs you about activities that may damage the equipment or other property. The following definition is based upon the ANSI Z535 definition.



NOTICE (blue): This indicates important information that is not related to personal injury which, if ignored, could result in damage to the equipment, damage to property, malfunctions, erroneous results, or loss of data.

2.2.5 Note

A Note is not a signal word and the text that is in this manual does not give safety information. A Note gives the reader useful supplementary information, hints, and tips.

2.3 General Protective Procedures



Important Safety Message

Make sure that all personnel who work on or near the equipment are capable of doing all operations in a safe way.

- Keep the manual in a convenient location near the equipment. Replace the manual if it becomes lost or damaged.
- Wear the applicable Personal Protective Equipment (PPE), in accordance with your plant's safety procedures.
- Understand the hazards of the equipment and the risks related to those hazards before working on or near the equipment.
- Obey all safety procedures of the local plant.
- Do not wear loose clothing, jewelry or long hair. Do not wear anything that can become entangled with the equipment.
- Be careful around the equipment to avoid hitting your head, arms, or other body parts against the equipment.
- Be careful not to trip over cables or other parts of the equipment.
- Do not move quickly in the area around the equipment.
- Do not climb, hang onto, or use any of the part of the equipment as a support
- Obey the lockout tagout (LOTO) procedures of the plant.
- If there is a safety-related malfunction when you are operating the equipment, press the emergency stop pushbutton. Inform your supervisor and follow the applicable steps approved by your company to fix the malfunction.

2.4 Safety Information for Various Activities



Important Safety Message

Read and understand all parts of the manual before using or working on any equipment.

The following sections list safety information for particular activities or groups of activities. Refer to the correct sections in the manual for more detailed instructions.

2.4.1 Transporting and Moving the Equipment

- Only qualified personnel are allowed to transport the equipment.
- Your company has sole responsibility for the safe moving and transporting of the equipment.
- Use safe moving procedures during transporting to maintain stability and to prevent the equipment from tipping or falling.
- Disconnect the equipment from the electrical power supply before moving.
- Use the correct lifting devices. If you use a forklift, lift the equipment at the correct lift points as shown by the blue lift point labels.
- If using a forklift, lift the equipment at the correct lift points as shown by the blue lift point labels.
- Blue lift point labels are placed on the equipment to show correct locations for lifting. These lift point locations were tested with the manufacturer's forklift trucks. A qualified rigger (material handler) must make sure that the lift points are correct for your lifting equipment.
- When you lift the equipment by hand, obey the safe lifting procedures of your company.

2.4.2 Installing

Only install the equipment if you have had the applicable training as defined by your company.

2.4.3 Operating the Equipment and Monitoring the Inspection Process

- Before beginning operation, make sure that the area is safe.
- Know the location and effect of each emergency stop button that controls the equipment.
- Do not operate the equipment without protective guards and doors in place.
- Do not reach into path of the products when any conveyors are in motion.
- Do not reach into the area around any sorting device, when the equipment is turned on.
- Make sure the safety circuit is working correctly.
- Do regular inspections of the equipment.
- If there is a fault or change in the equipment behavior, stop the equipment and inform responsible personnel.
- Only operate the equipment if you have the applicable training as defined by your employer.

2.4.4 Testing and Verifying the Equipment

Only test and verify the equipment if you have had the applicable training as defined by your company.

2.4.5 Maintaining, Cleaning and Sanitizing the Equipment

- Remove all power from the equipment before doing any work.
- Keep the equipment in good working order.
- Follow a preventative maintenance program.
- Replace parts when needed.
- Obey the lockout tagout (LOTO) procedures of the plant.
- Test (validate) the safety circuit after parts are replaced.
- Only use METTLER TOLEDO approved spare parts and accessories.
- Do not make any modifications to the equipment without official authorization from METTLER TOLEDO.
- Replace safety labels if damaged, missing, or unreadable.
- Do a visual check of the equipment at least once during a shift to identify any visual damage or faults.
- Report any equipment changes immediately to the responsible supervisor.
- When required for a hygienic production environment, do regular sanitizing of the equipment according to your company's procedures.
- After cleaning or sanitizing, check all cables connectors and compressed air hoses for leakage, loose connections, rub marks and damage. Tighten, repair or replace any faulty cables and air tubing, as necessary.

2.5 Special Hazards

The following sections describe special instructions for equipment that may have special hazards.

2.5.1 Electricity

- Only work on the electrical systems if you have the applicable electrical training as defined by your company.
- Keep all electrical enclosure doors closed. If the doors have locks on them, keep them locked.
- Remove all power from the equipment before doing any work.

2.5.2 Pneumatics

- Only work on the pneumatic system if you have the applicable training as defined by your company.
- Check pneumatic parts for visible damage.
- Before any repair or maintenance work, relieve the pressure of compressed air pipes, tubing or pneumatic system parts.
- Make sure that there is a consistent compressed air supply when you operate the equipment.
- If a pneumatic pipe becomes loose during the operation of the equipment, keep away from the loose pipe.

2.5.3 Moving Parts and Hot Parts

- Product inspection equipment moves automatically, intermittently and without notice. Some surfaces may get hot. Do not place hands or any other body parts on or near the moving parts.
The following are examples of moving parts and hot parts:

- Conveyor belts and rollers
 - Rejectors and product diverters
 - Motors
 - Gearboxes
 - Side grips
 - Automatic position adjusters
 - Encoders
 - Pneumatic cylinders
 - Solenoids
- Only operate the equipment with all safeguards in place.
 - Make sure that all safeguards are in correct operating condition.
 - Do not climb, sit, stand, walk or ride on the conveyor.

2.5.4 Rejectors

- Be aware around rejectors because they move quickly and without notice.
- Do not reach into the rejector area or reject bin while the rejector is operating.
- Make sure the rejector is installed correctly.
- Make sure the rejected products do not cause harm.
- Keep your eyes and other body parts away from air blast nozzles when the equipment is operating.

2.5.5 Noise

- Make sure that all noise-absorbing parts of the equipment are present and installed in the correct position.
- Wear hearing protection as directed by your local safety authority.

2.5.6 Electromagnetic Fields

- High frequency electromagnetic fields exist in the aperture of the metal detector and in the area near the aperture.
- When the metal detector is operating, avoid prolonged exposure to the electromagnetic field inside the aperture and up to 200 mm from the aperture.
- Consult your doctor if you have a pacemaker or similar device before you operate the metal detection system or approach the metal detector's aperture.

2.5.7 Lights and Laser Sensors

- Do not stare directly at any lights or lasers.
- Avoid prolonged exposure to ultraviolet (UV) light, infrared (IR) light, and lasers.

2.5.8 Strobe Lights

Strobe lights can cause seizures in individuals with photosensitive epilepsy.

- Individuals with photosensitive epilepsy must not operate the equipment.
- Do not operate the equipment when excessively fatigued or after consuming alcohol.
- Do not look directly at the lights, especially at close distances.
- Avoid placing the equipment in areas with reduced lighting.
 - If lights are inside of an enclosure, do not open the enclosure doors when lights are flashing.

2.5.9 X-ray Radiation

- The equipment meets international and local requirements for radiation exposure with less than or equal to 1 $\mu\text{Sv/h}$ emission at a distance of less than or equal to 10 cm from any external surface. USA and Canada have adopted 5 $\mu\text{Sv/h}$ as their maximum emission value.
- METTLER TOLEDO tests the equipment for radiation emissions before shipment and records the results in a Critical Examination Report.
- Do not operate the equipment with any part of the body within the conveyor tunnel ends or within the protective radiation shielding curtains
- Do not operate the equipment if the protective radiation shielding curtains are damaged. Check the curtains regularly and replace them if necessary.
- Test the equipment when it is installed to make sure it is below the safe level for radiation emission. Install additional shielding if necessary as directed in the technical manual supplied with the equipment.
- It is the end user's responsibility to observe the legislation and local rules with regards to operating X-ray equipment.

2.5.10 Explosive Atmospheres or Locations

- Use only properly evaluated and marked equipment in a potentially explosive atmosphere or location.
- Requirements for marking equipment vary by jurisdiction, and you must identify the regulations and standards that apply. For example, the ATEX directive (2014/34/EU) with EN 60079-0 apply to the European Union, and the Code of Federal Regulations (CFR 1910.307) with NFPA 70 apply to the USA.
- Only qualified persons may classify a potentially explosive atmosphere or location
- If your equipment is marked for use in a potentially explosive atmosphere or location, read the additional information provided before installation.

3 Design and Function

3.1 Options for Metal Detection Systems

This section gives a list of the standard components that are installed on your system together with the most common optional components that can be installed.

Stop-on-Detect Systems - Standard and Optional Components

Standard components	Optional components
Metal detector	Warning beacon stack
Conveyor belt (flat polyurethane (PU) or modular)	Audible warning device
Adjustable height feet	Adjustable height lockable casters
Industrial control enclosure	Infeed pack sensor
Metal detector power supply enclosure	Conveyor speed sensor
Motor with gearbox unit	Conveyor speed control
	Reset key switch or pushbutton
	Product guides

Automatic Reject Systems - Standard and Optional Components

Standard components	Optional components
Metal detector	Warning beacon stack
Audible warning device	Adjustable height lockable casters
Conveyor belt (flat PU or modular)	Conveyor speed sensor
Industrial control enclosure	Conveyor speed control
Metal detector power supply enclosure	Reject bin (various types and locations)
Adjustable height feet	Bin full sensor
Motor with gearbox unit	Reject Confirmation Unit (RCU) sensor
Automatic reject mechanism	Infeed pack sensor
Air pressure switch	Reset key switch or pushbutton
Electrically-operated shutoff/dump valve	Product guides
Reject bin lock - mechanical or electro-mechanical	

Due Diligence Systems - Standard and Optional Components

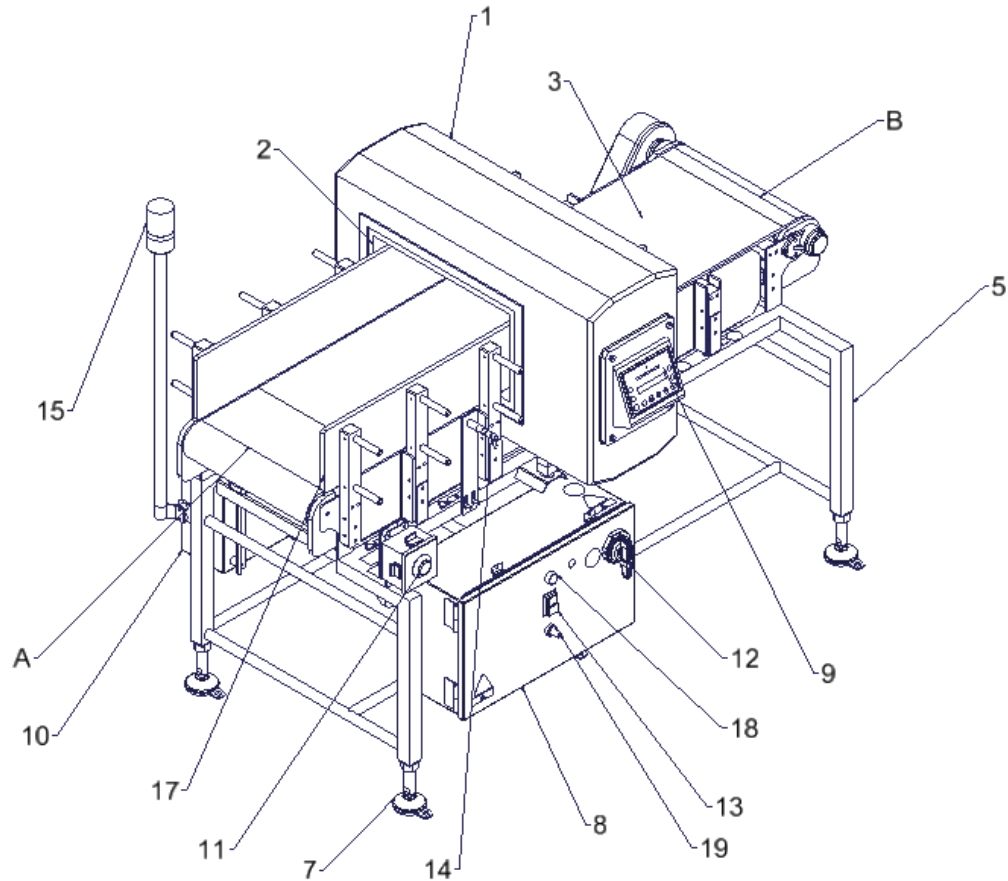
Standard components	Optional components
Metal detector	Adjustable height lockable casters
Audible warning device	Product guides
Adjustable height feet	Conveyor speed control
Conveyor belt (flat polyurethane (PU) or modular)	Warning beacon stack
Industrial control enclosure	
Metal detector power supply enclosure	
Motor with gearbox unit	
Infeed pack sensor	
Automatic reject mechanism	
Air pressure switch	
Conveyor speed sensor	
Electrically-operated shutoff/dump valve	
Bin full sensor	
Reject Confirmation Unit (RCU) sensor	
Reset key switch or pushbutton	
Reject check sensor	
Reject bin lock - mechanical or electro-mechanical	

3.2 Components of a Stop-on-Detect System

This section describes the components that are installed on a typical stop-on-detect system.

The stop-on-detect system does not have an automatic reject mechanism. When the system detects a contaminated product, it stops the conveyor belt and it may give the user a visual or audible alarm.

NOTE: All of the components that are installed on a typical stop-on-detect system are also installed on an automatic reject system and a due diligence system.



Item	Description	Item	Description
A	Conveyor infeed	10	Metal detector power supply
B	Conveyor outfeed	11	Emergency stop button
1	Metal detector	12	Disconnect (on control panel or Industrial control enclosure)
2	Metal detector aperture	13	Stop/start conveyor buttons
3	Conveyor belt (flat polyurethane (PU) or modular)	14	Infeed pack sensor (optional)
4	Motor with gearbox unit	15	Warning beacon (optional)
5	Welded stainless steel frame	16	Audible alarm (optional)
6	Adjustable height lockable casters (optional)	17	Product guides
7	Adjustable height feet	18	Reset key switch or pushbutton (optional)
8	Industrial control enclosure	19	Conveyor speed control (optional)
9	HMI screen		

All of the systems have a metal detector (1) that consists of a metal casing with a rectangular aperture (2). The detector (1) inspects every product for contamination by ferrous metals, non-ferrous metals and stainless steel particles. If the detector (1) identifies a contaminated product, it stops the conveyor belt and gives the user a visual signal on the HMI screen (9).

NOTE: Please refer to the applicable manual for instructions about operating the metal detector installed on your system.

The metal detector (1) is installed on a one-piece welded stainless steel frame (5) that has adjustable height feet (7) or adjustable height lockable casters.

A flat PU or modular conveyor belt (3) transports products at a constant rate from the conveyor infeed (A) through the metal detector aperture (2) until the products reach the conveyor outfeed (B).

A motor with gearbox unit (4) drives the conveyor belt (3). The metal detector (1) inspects the products to identify any unwanted metal contaminants.

There are two electrical enclosures installed on the metal frame. One enclosure is the industrial control enclosure (8) and the other enclosure is the metal detector power supply (10).

The industrial control enclosure (8) is installed at a convenient location so that the user can start and stop the conveyor belt and easily access the emergency stop button (11).

If there is an emergency, press the emergency stop button (11) to stop the conveyor belt. A disconnect (12) is installed on the industrial control enclosure (8). The disconnect allows service personnel to isolate the electrical power supply to the system to do maintenance and cleaning.

One of these two types of product guides (17) can be installed on the conveyor:

- Along the conveyor infeed (A) (as shown in the figure above)
- Along the full length of the conveyor belt

NOTE: Installation personnel can easily adjust the positions of the product guides (17) so that they match the width of different sizes of products. This is only for optional adjustable product guides.

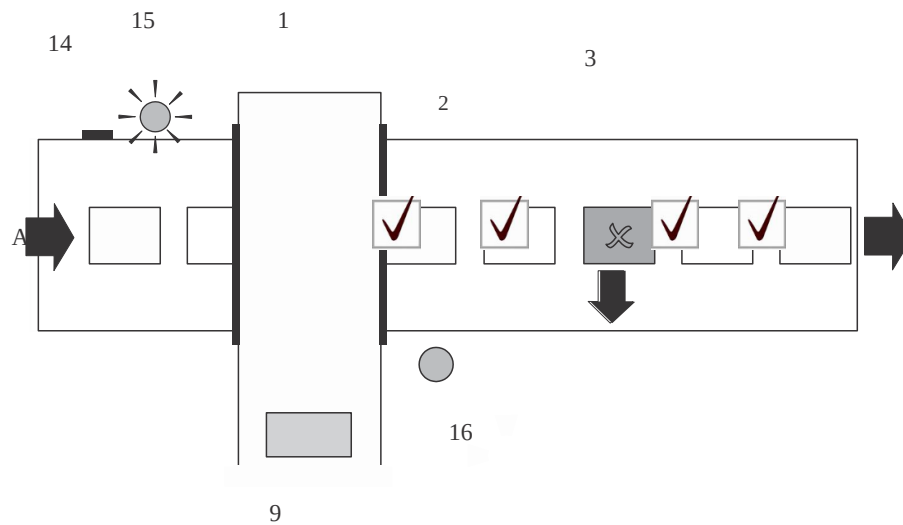
A stop-on-detect system may use a single warning beacon (15). The conveyor belt (3) stops and the beacon (15) illuminates when the metal detector (1) identifies a contaminated product.

An audible alarm (16) may be installed in addition to, or instead of, the warning beacon (15). The alarm sounds when the metal detector (1) identifies a contaminated product.

3.3 How the Stop-on-Detect System Works

This section describes how the stop-on-detect system works. The aim of this section is to give new users a basic understanding of the system before they operate it.

NOTE: Refer to 'Components of a Stop-on-Detect System' for a description of the components. A system without an automated reject mechanism is called a stop-on-detect system.



A conveyor, that is located before the conveyor infeed (A), automatically feeds products onto the moving conveyor belt (3). Alternatively, a user manually feeds products onto the conveyor infeed (A).

If installed, the infeed pack sensor (14) counts the products and sends a signal to the metal detector (1) to record the position of the products on the conveyor belt (3) in relation to the metal detector aperture (2)

The conveyor belt (3) transports the products through the aperture (2) of the metal detector (1) where the detector examines them for metal contamination.

If a product **is not contaminated**, the system allows it to travel along the full length of the conveyor belt (3). An outfeed conveyor automatically carries the product away from the metal detection system, or a user manually removes the product.

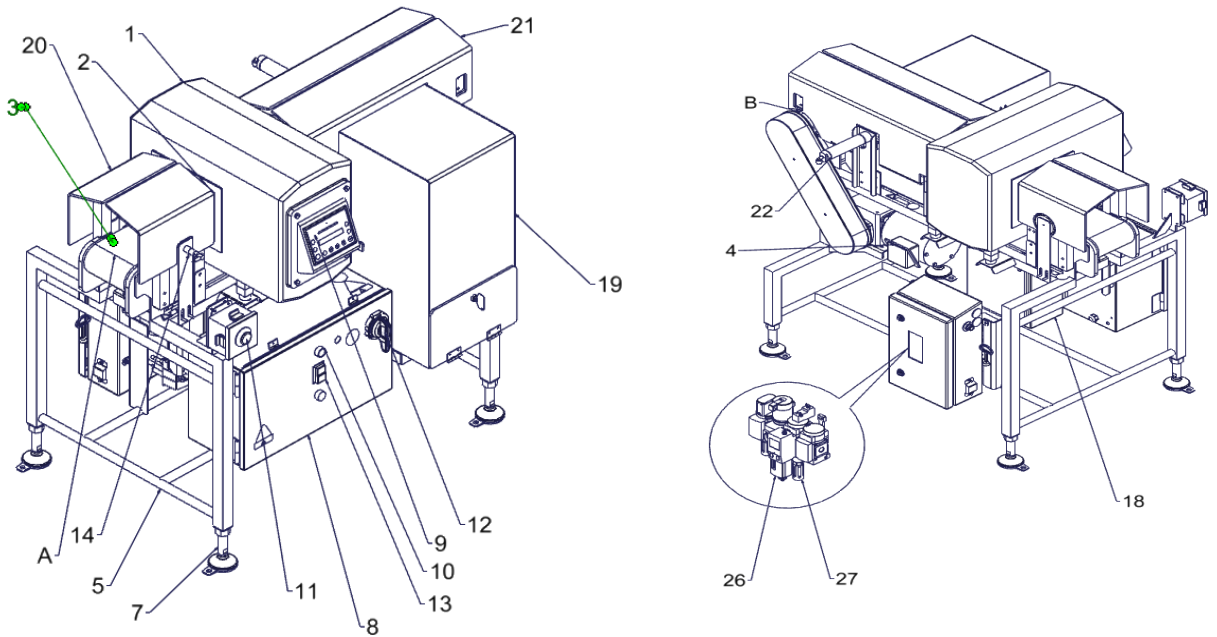
If a product **is contaminated**, the metal detector (1) sends a signal to stop the conveyor belt (3). If installed, the warning beacon (15) illuminates and/or the audible alarm (16) sounds. These give the user a visual and/or audible warning when the system has identified a contaminated product.

After the conveyor belt (3) has stopped, the user can manually remove the contaminated product from the production line.

If there are a large quantity of small products on the conveyor belt (3) when it stops, the user needs to remove all of the products from the belt.

3.4 Additional Components of an Automatic Reject System

This section describes the components that are installed on a typical automatic reject system. The components (with bold numbers on the figure below) are in addition to the components installed on a typical stop-on-detect system (refer to 'Components of a Stop-on-Detect System').



Basic Components of All Systems

Item	Description	Item	Description
A	Conveyor infeed	9	HMI screen
B	Conveyor outfeed	10	RESET key switch or pushbutton
1	Metal detector	11	Emergency stop button
2	Metal detector aperture	12	Disconnects (on Industrial control enclosure)
3	Conveyor belt (flat PU or modular)	13	Stop/start conveyor buttons
4	Motor with gearbox unit	14	Infeed pack sensor (optional)
5	Welded stainless steel frame	15	Warning beacon stack (optional)
6	Adjustable height lockable casters (optional)	16	Audible alarm (optional)
7	Adjustable height feet	17	Product guides
8	Industrial control enclosure	18	Metal detector power supply

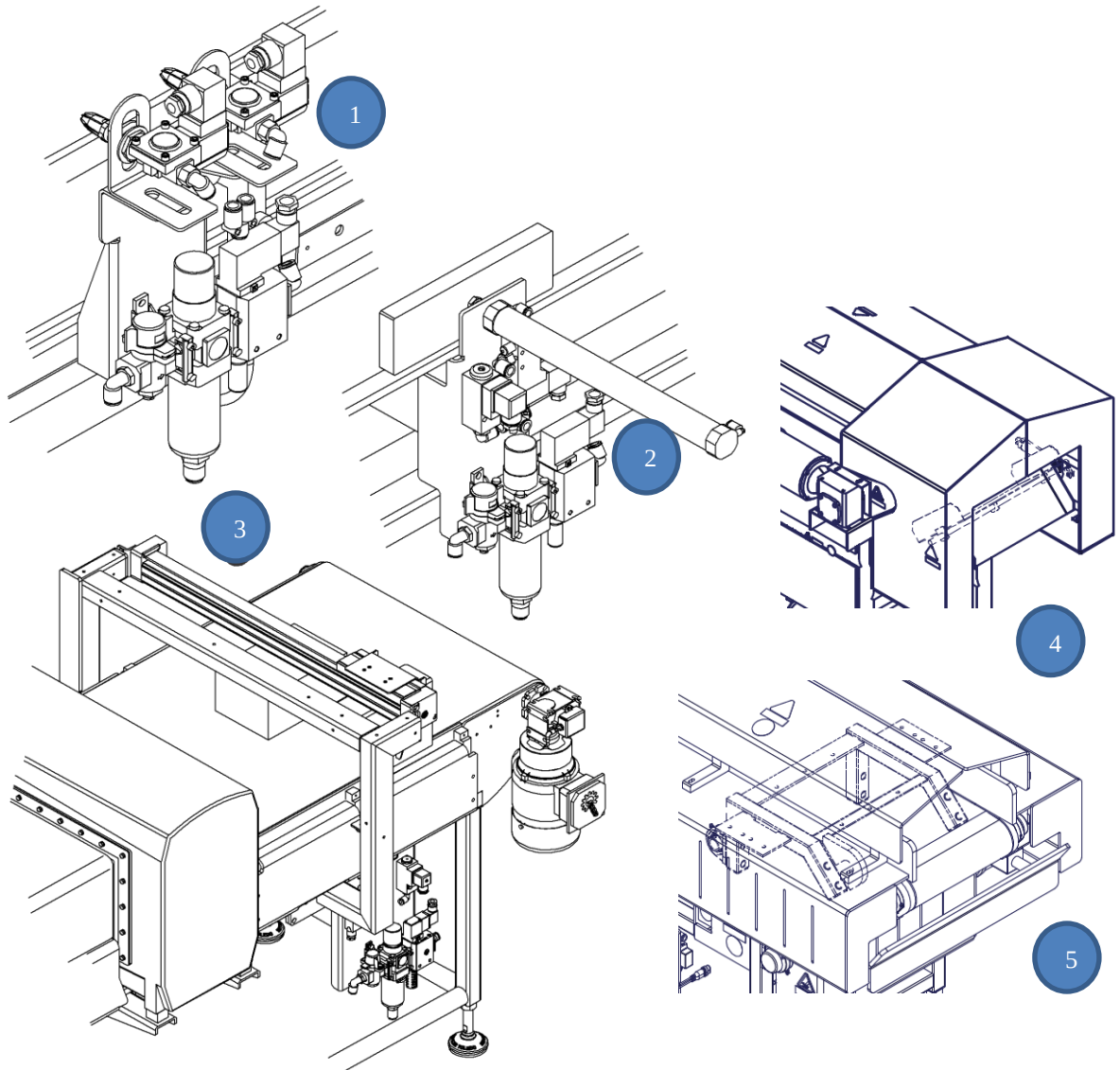
Additional Components of an Automatic Reject System

Item	Description	Item	Description
19	Lockable reject bin	24	Bin full sensor (optional)
20	Infeed tunnel guard (optional-not shown)	25	Reject Confirmation Unit (RCU) sensor (optional)
21	Outfeed tunnel guard(optional)	26	Electrically-operated pneumatic dump valve
22	Automatic reject mechanism	27	Air pressure switch
23	Conveyor speed sensor with conveyor speed control (optional)		

Automatic Reject Mechanism

The metal detector activates the automatic reject mechanism (22) when the detector identifies metal contamination in a product that is placed on the conveyor belt (3). The purpose of the reject mechanism (22) is to remove contaminated products from the production line.

Many types of reject mechanisms are available (see examples in the figure below). All are pneumatically operated such as air blast mechanisms (1), pushers and punches (2), overhead sweeps (3) end flap rejector (4) retracting band reject (5).



Warning Beacon Stack

An automatic reject system can have up to five high-visibility and color-coded warning beacons (15). The beacons are joined together on top of each other in a stack. The stack is installed on the welded metal frame to give the user a visual warning of the following (but not limited to) types of hazards and metal detection system faults as they occur when the system is operating:

- Metal detector fault
- Metal contamination detected
- Reject Confirmation Unit (RCU) fault
- Pneumatic system air failure
- Reject bin full

Tunnel Guards

An infeed tunnel guard (20) and an outfeed tunnel guard (21) may be installed. These guards prevent access to moving reject system parts and that make sure that users cannot remove contaminated products from the production line.

Air Pressure Switch

The air pressure switch (27) monitors the air pressure that supplies the automatic reject mechanism (22). If there is an air leak in the reject mechanism (22), or low air pressure in the pneumatic system, the air pressure switch (27) operates. This stops the conveyor belt (3), and it activates a warning beacon (15), an audible alarm (16), or both of these devices.

Electrically-Operated Pneumatic Dump Valve

The electrically-operated pneumatic dump valve operates to exhaust all of the air from the pneumatic system when someone presses the emergency stop button.

Audible Alarm

An audible alarm (16) can be installed in addition to or instead of the warning beacon (15). The alarm sounds when the metal detector (1) identifies a contaminated product.

Lockable Reject Bin

The lockable reject bin (19) gives a secure and temporary storage area for rejected products. An authorized person or a nominated key holder locks the reject bin to make sure that users cannot remove contaminated products from the bin and reintroduce them back into the production line.

A manually-operated lock or an electro-mechanical lock is provided on the reject bin so that only authorized personnel can remove contaminated products from the inspection system. An authorized person can open the electro-mechanical lock only by activating a command on the screen (9) of the metal detector (1).

Infeed Pack Sensor

The infeed pack sensor (14) identifies when a product is fed onto the conveyor infeed (A). The sensor works in conjunction with the metal detector (1) and the automatic reject mechanism (22). The metal detector (1) uses the signal from the infeed pack sensor (14) to calculate the exact position of a contaminated product on the conveyor belt (3). This allows the metal detector (1) to activate the automatic reject mechanism (22) to remove the product from the production line.

Conveyor Speed Sensor with Conveyor Speed Control

The conveyor speed sensor (23) monitors the speed of the motor (4) and converts this into the speed of the conveyor belt (3). The conveyor speed sensor (23) sends a signal to the metal detector (1) so that the detector can calculate the position of the products on the conveyor belt (3) as the system transports them along the belt.

The following devices work together to make sure that there is accurate rejection of contaminated products by the inspection system:

- Infeed pack sensor (14)
- In-built reject timer (located in the metal detector)
- Conveyor speed sensor (23)

Reject Confirmation Unit (RCU) Sensor

A Reject Confirmation Unit (RCU) (24) optical sensor is located across the entrance of the reject bin (19). As soon as the metal detector (1) identifies a contaminated product, the system is configured to expect a follow-on signal from the RCU (24) to confirm that the contaminated product has entered the reject bin (19). If the system does not receive this confirmation signal, the system raises an alarm and stops the conveyor belt (3).

Reset Key Switch or Pushbutton

All of the elements that result in following actions are linked to a manually-operated reset key switch (18) or pushbutton.

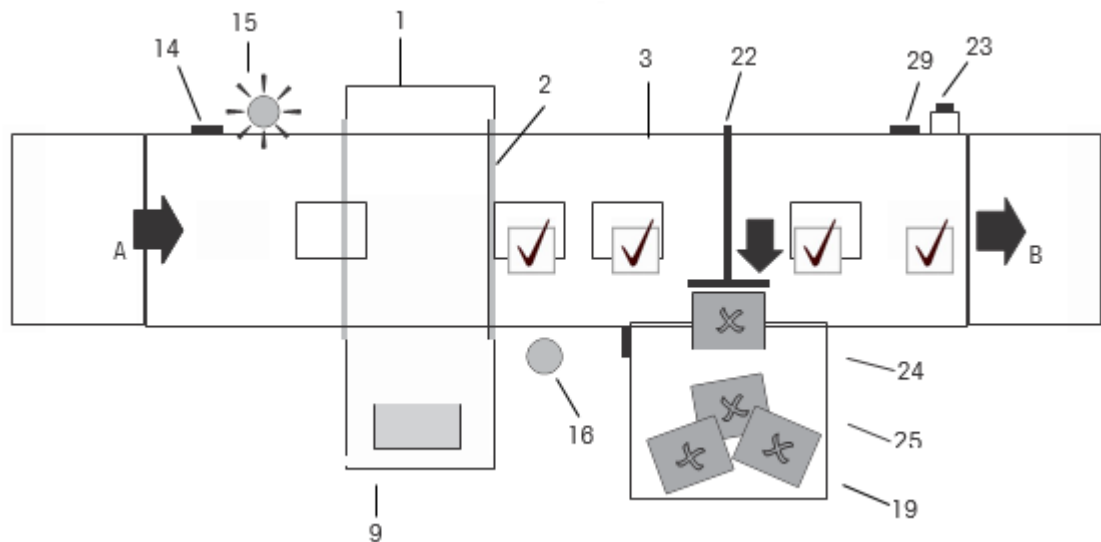
- Stopping the conveyor belt (3)
- Resetting the Reject Confirmation Unit (RCU) fault condition (24)
- Resetting the warning beacons (15) and the audible alarm (16), if installed

Only authorized personnel or nominated key holders are allowed to restart the system after a fault, or after the conditions that caused the fault, have been sorted out.

3.5 How the Automatic Reject System Works

This section describes how the automatic reject system works. The aim of this section is to give new users a basic understanding of the system before they operate it.

NOTE: Refer to 'Components of an Automatic Reject System' for a description of the components.



The conveyor belt (3) transports the products through the aperture (2) of the metal detector (1) where the detector examines them for metal contamination.

If a product is **not contaminated**, the system allows it to travel along the full length of the conveyor belt (3). An outfeed conveyor automatically carries the product away from the metal detection system.

If a product **is contaminated**, the metal detector (1) activates the automatic reject mechanism (22) so that it rejects the product from the production line into a reject bin (19).

An authorized person opens the reject bin (19) and removes the contaminated product.

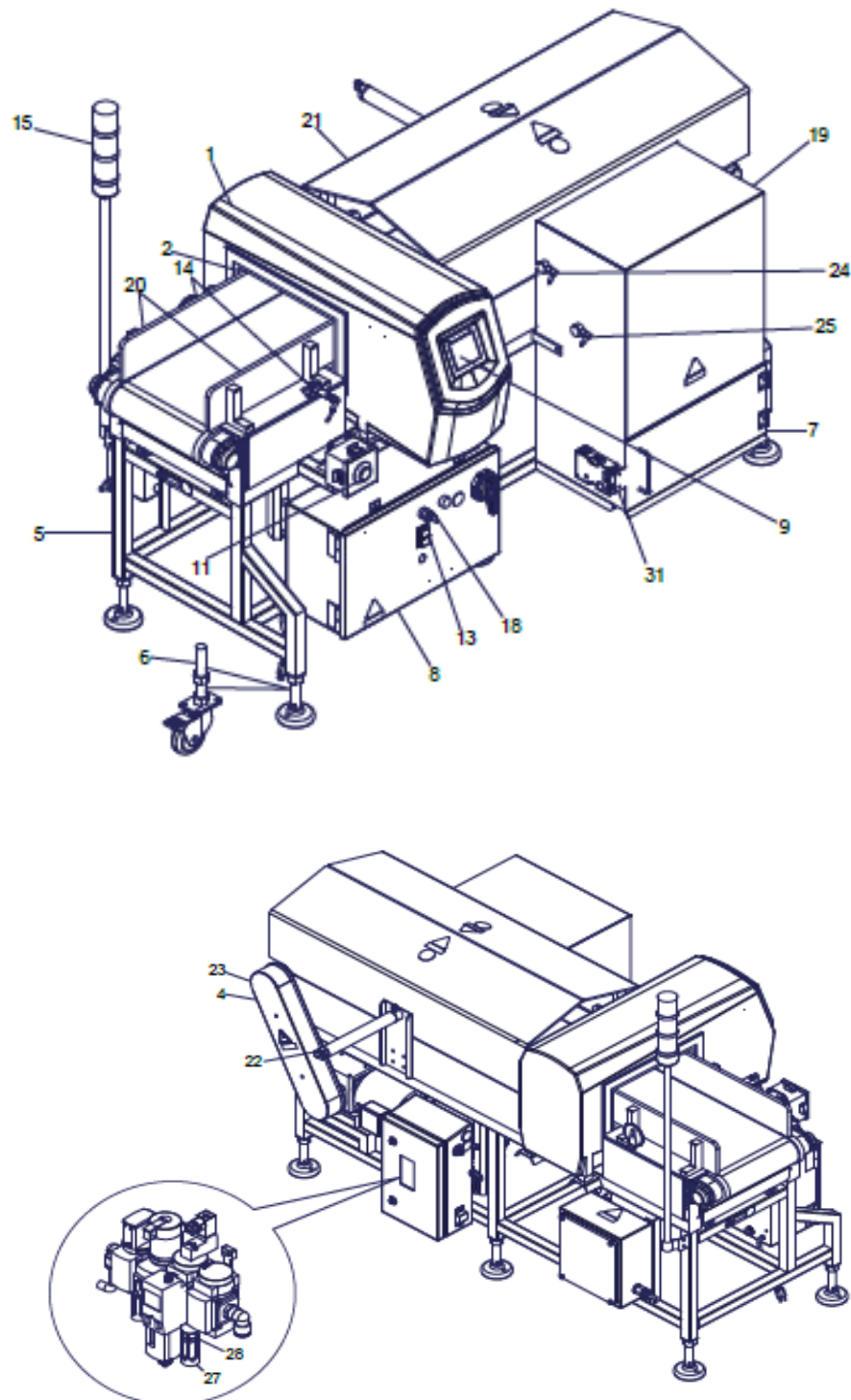
A warning beacon stack (15), an audible alarm (16), or both of these devices can be installed on the system. These give the user a visual and/or audible warning when the system has identified a contaminated product. They also give an indication of known faults and warnings in the metal detection system.

NOTE: Only authorized personnel can access settings on the HMI screen (9) to reset the visual and audible warning devices.

3.6 Additional Components of a Due Diligence System

This section describes the components that are installed on a due diligence system. The components (shown on the figure below) are in addition to the components installed on a typical stop-on-detect system (refer to 'Components of a Stop-on-Detect System').

The components of the due diligence system work together to make sure that the system identifies any contaminated products and removes them automatically from the production line.



Basic Components of All Systems

Item	Description	Item	Description
A	Conveyor infeed	9	HMI Screen
B	Conveyor outfeed	10	Metal detector power supply
1	Metal detector	11	Emergency stop button
2	Metal detector aperture	12	Disconnect switch (on control panel or Industrial control enclosure)
3	Conveyor belt (flat polyurethane (PU) or modular)	13	Stop/start conveyor buttons
4	Motor with gearbox unit	14	Infeed pack sensor (optional)
5	Welded stainless steel frame	15	Warning beacon stack (optional)
6	Adjustable height lockable casters (optional)	16	Audible alarm (optional)
7	Adjustable height feet	17	Product guides
8	Industrial control enclosure	18	Reset key switch or pushbutton(optional)

Additional Components of a Due Diligence System

Item	Description	Item	Description
19	Lockable reject bin	26	Reset key switch or pushbutton
20	Infeed tunnel guard	27	Air pressure switch
21	Outfeed tunnel guard	28	Electrically-operated pneumatic dump valve
22	Automatic reject mechanism	29	Reject check sensor (fitted on level 3 due diligence system)
23	Conveyor speed sensor with conveyor speed sensorl (optional on level 1, 2 and 3 due diligence systems and fitted on all variable speed systems)	31	Reject bin electro-mechanical lock (optional on level 1, 2 and 3 due diligence systems)
24	Reject Confirmation Unit (RCU) sensor		
25	Bin full sensor		

Reset Key Switch or Pushbutton

All of the elements that result in following actions are linked to a manually-operated system reset key switch (18) or pushbutton.

- Stopping the conveyor belt (3)
- Activating the key switch or pushbutton (26) to reset the electrically-operated pneumatic dump valve (28).
- Triggering the Reject Confirmation Unit (RCU) fault condition (24)
- Resetting the metal detector (1)
- Activating the warning beacons (15) and the audible alarm (16), if installed

Only authorized personnel or nominated key holders are allowed to restart the system after a fault, or after the conditions that caused the fault, have been sorted out.

Reject Inhibit Key Switch or Pushbutton

You use the Reject Inhibit key switch or pushbutton (30) to test that the Reject Confirmation Unit (RCU) sensor (24) is working correctly before you run products through the system. Refer to section 'Testing the Reject Confirmation Unit (RCU) sensor'.

Reject Check (Outfeed) Sensor

The reject check sensor (29) is a device that checks if the due diligence system has successfully rejected a contaminated product or not.

The reject check sensor (29) works in conjunction with the infeed pack sensor (14), the Reject Confirmation Unit (RCU) sensor (24) and the metal detector (1).

The automatic reject mechanism (22) can have rejected a contaminated product and the Reject Confirmation Unit (RCU) sensor (24) can have notified the metal detector (1) that the product has entered the reject bin (19). It is unlikely that the rejected product will come out of the reject bin (19) and travel down the conveyor belt (3) to the reject check sensor (29). However, if this happens, the reject check sensor (29) identifies that the contaminated product is still on the production line so it triggers an alarm to stop the conveyor belt (3).

Reject Bin Electro-Mechanical Lock

If an electro-mechanical lock (31) is installed on the reject bin (19), an authorized person can enter a code on the screen (9) to unlock it. They do this to remove rejected products from the reject bin (19) while the conveyor belt is running.

The authorized person needs to remove the rejected products within a time limit that is preset on the system, otherwise the system stops the conveyor belt (3).

NOTE: If the electro-mechanical lock (31) is not properly closed, the system stops the conveyor belt (3).

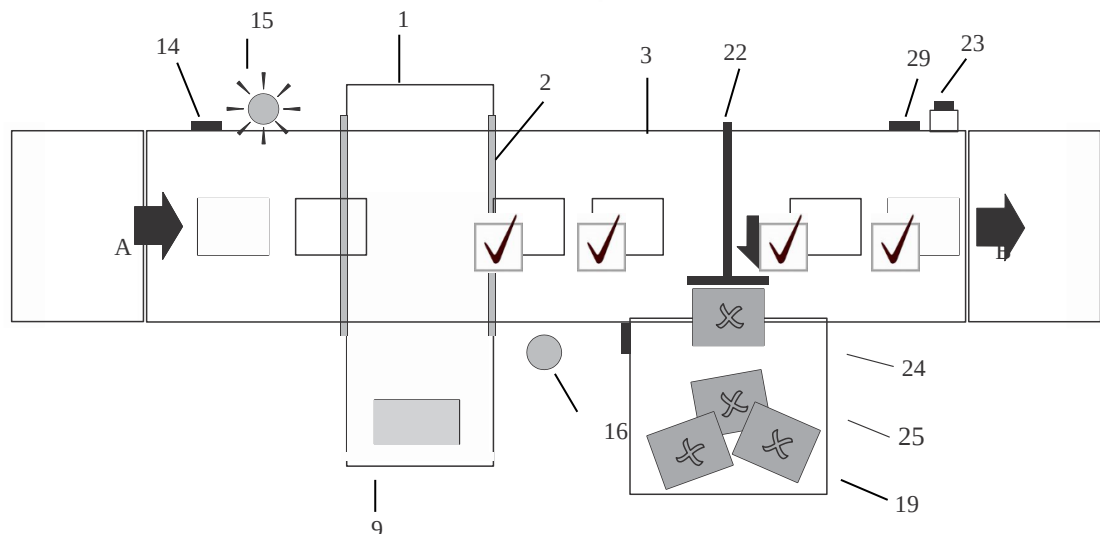
3.7 How the Due Diligence System Works

This section describes how the due diligence system works. The aim of this section is to give new users a basic understanding of the system before they operate it.

NOTE: Refer to 'Components of a Due Diligence System' for a description of the components.

A conveyor that precedes the conveyor infeed (A) automatically feeds products onto the conveyor belt (3), making sure that the products are evenly-spaced. The moving products break the light beam on the infeed pack sensor (14). The metal detector (1) uses the signal from the infeed pack sensor (14), together with information from the conveyor speed sensor (23), to calculate the position of the pack on the conveyor belt (3) in relation to the following due diligence components:

- Metal detector (1)
- Automatic reject mechanism (22)
- Reject Confirmation Unit (RCU) sensor (24)
- Reject check sensor (29)



The above components of the automated due diligence system work together to make sure that the metal detection system identifies any contaminated products and removes them automatically from the inspection process of the production line.

The conveyor belt (3) transports the products through the aperture of the metal detector (2) where the detector (1) examines them for metal contamination.

If a product **is not contaminated**, the due diligence system allows it to travel along the full length of the conveyor (3) until an outfeed conveyor automatically carries the product away from the metal detection system.

If a product **is contaminated**, the metal detector (1) activates the automatic reject mechanism (22) so that it rejects the product from the inspection system into a lockable reject bin (19).

A Reject Confirmation Unit (RCU) sensor (24) is located across the entrance of the reject bin (19). This RCU registers that the contaminated product has entered the reject bin (19).

A bin full sensor (25) sends a signal to the metal detector (1) that warns the user and stops the conveyor (3) when the reject bin (19) is about 80% full of contaminated products. An authorized person unlocks the reject bin (19) and removes the contaminated products.

A warning beacon stack (15), an audible alarm (16), or both of these devices may be installed on the system. These give the user a visual or audible warning when the system has identified a contaminated product as well as an indication about hazards and faults in the metal detection system.

NOTE: Only authorized personnel can access settings on the screen (9) to reset the visual and audible warning devices

4 Installation

4.1 Transporting and Unpacking the System

Depending on the location of the customer, METTLER TOLEDO supplies the system installed on a wooden pallet or in a wooden or cardboard crate installed on a wooden pallet.

Follow these step to transport and unpack the system:



DANGER

If you want to move the crate using a forklift truck, pick up and move the crate with the side of the crate that shows the shipping label or text that says "LIFT THIS SIDE ONLY" (or similar text) facing towards the front of the forklift truck.

- 1 If applicable, carefully remove and recycle or discard the four side panels of the crate.
- 2 If applicable, carefully remove and discard the packaging and protective materials inside the crate.
- 3 If your system has lockable casters installed, remove and discard the blocks and metal brackets that attach the casters to the pallet.
- 4 If your system has adjustable height feet installed, remove and discard the metal brackets that attach the feet to the pallet.

4.2 Lifting and Moving the System

Whenever possible, use a forklift truck to lift and move the system. Alternatively, use powered lifting equipment with straps or lifting tackle.

Allow enough space at the front and the back of the system so that commissioning and maintenance personnel can easily get access to components. Whenever possible, allow a minimum 1 meter of free space at the front of the system and 1 meter at the back of the system.



DANGER

Lift the system only under the supervision of a competent person who has sufficient and applicable training.



DANGER

Obey the applicable safety regulations at all times during lifting operations.



NOTICE

The manufacturer accepts no liability for damage to the system caused by using incorrect lifting procedures or inadequate lifting equipment.

4.2.1 Lifting the System Using a Forklift Truck



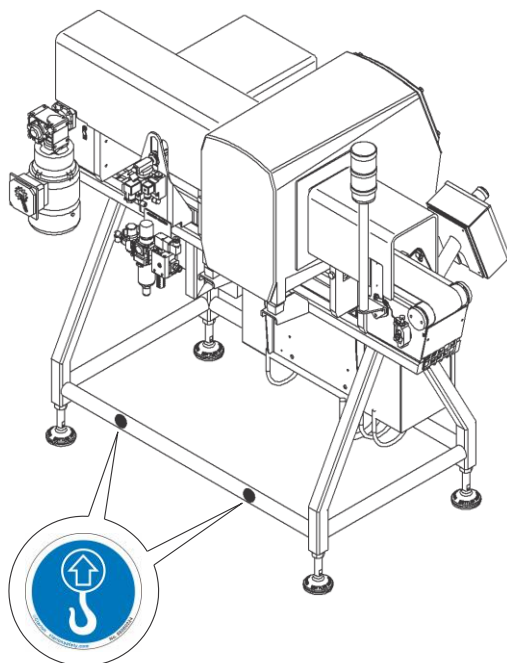
DANGER

Some metal detection systems have a high center of gravity so they can be top-heavy. Take care when you use a forklift truck to move the system on a slope as the system can topple over.

If the system topples over and falls off the forklift truck, it can result in death, severe injury to personnel or damage the equipment.

Follow these steps to lift and move the metal detection system using a forklift truck:

- 1 Position the forks of the truck immediately below the two lifting points **with the two lifting point labels facing the front of the forklift truck**. **NOTE:** Two labels that show the lifting points are attached to one of the lower cross members of the welded metal frame (see figure below).
- 2 If the forks on your truck are wider or narrower than the positions of the lifting point labels, position the forks at an even distance from the positions of the two labels.
- 3 Raise the system gradually until you have raised it enough to make sure that your forklift truck has found the center of gravity for the system and that you are certain about lifting the system.
- 4 If necessary, lower the forks and reposition the forks to a position that is correctly balanced and in the exact center of gravity of the system.
- 5 Position the system on the forks so that it does not topple over and fall away from the forklift truck.
- 6 Raise the system gradually until you have lifted it enough to clear the ground and any obstacles.
- 7 Move the system carefully to the correct location.
- 8 Lower the forks gradually and withdraw them from the underside of the system.



4.2.2 Lifting the System Using Powered Lifting Equipment



WARNING

Always use the correct lifting equipment to move the metal detection system. Inspect all of the lifting straps and lifting tackle before you lift the system and make sure that you do not exceed the safe working load. Do not use straps with frayed or broken strands and do not use lifting tackle with damaged hooks and rings.



DANGER

Always lift the system in a safe way using applicable slinging methods and by using powered lifting equipment.

If you cannot lift and move the metal detection system using a forklift truck, you can follow these steps to lift and move the system using powered lifting equipment:



NOTICE

Do not pass lifting straps through the aperture of the metal detector or around the skid plate to lift the system.

Ignoring this notice will cause damage to the metal detector and the metal detection system.

- 1 Position the lifting straps or the lifting tackle under the upper welded metal frame of the metal detection system so that the system will not tilt when you lift it.
- 2 Attach the straps to the powered lifting equipment.
- 3 Raise the system gradually and move it to the correct location.
- 4 Lower the system gradually.
- 5 Remove the lifting straps or the lifting tackle.

4.3 Positioning the System



NOTICE

To move a system that has casters installed, push on the metal frame only. Do not move the system by pressing on any other components such as the reject bin, tunnel guards, product guides, and so on, as this can cause damage to the system.



NOTICE

Do not push on the metal detector to move the system as it is heavy and this can cause the system to topple over.

Only use casters provided by Mettler Toledo.

If the system has lockable casters installed, you can unlock the casters and move the system carefully on the wheels by pushing on the welded frame. You can also move and position the system on different process lines. Avoid hitting any objects and obstacles as this can damage the wheels.

The system is supplied for installing onto a floor, with or without lockable casters, depending on the customer's requirements.

NOTE: Provide sufficient power and pneumatic supplies at all of the working positions. Follow these points when you position the metal detection system:

- Avoid supporting the system on or near vibrating structures or near vibrating machines.
- Position the system so that servicing personnel have easy access to it.
- Do not position any other cables that carry switched or heavy load next to the metal detector cable. This can cause electrical interference.

4.4 Adjusting the Height of the System

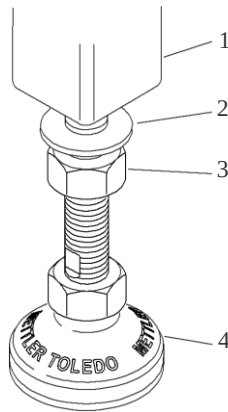
This section is aimed at the installation technician. It describes how to adjust the height of the infeed and outfeed ends of the system and how to level the system. This is important because the preceding conveyor need to transfer products easily onto the infeed roller of the conveyor. Also, the outfeed roller needs to transfer products easily off the conveyor outfeed.

4.4.1 Leveling the Conveyor

If your metal detection system has adjustable feet or adjustable and lockable casters installed, you can level the conveyor. To do this, you put a long spirit/bubble level diagonally along the length of the conveyor and then adjust the feet or casters, as necessary, to level the system.

4.4.2 Adjusting the Feet

If your system has adjustable height feet installed, you can level the system and adjust the infeed and the outfeed heights of the conveyor by raising and lowering the feet.



Item	Description
1	Leg of the welded metal frame
2	Flat washer
3	Locknut
4	Adjustable height foot

Follow these steps to adjust the height of the feet:

- 1 Slacken the locknut (3).
- 2 Use a spanner/wrench to rotate the adjustable height foot (4) in a clockwise direction, as necessary, to reduce the height of the frame (1).
- 3 Use a spanner/wrench to rotate the adjustable height foot (4) in a counterclockwise direction, as necessary, to increase the height of the frame (1).
- 4 When you have achieved the correct height for the adjustable foot (4), turn the locknut (3) in a clockwise direction until it locks the adjustable height foot (4) tightly in the frame (1).
- 5 Adjust all of the adjustable height feet (4) until the welded metal frame is positioned firmly on the ground.
- 6 Make sure that the adjustable height feet (4) evenly support the welded metal frame and the metal detection system so that it does not rock when the system operates.
- 7 Apply silicone sealant around the edge of the feet and floor.

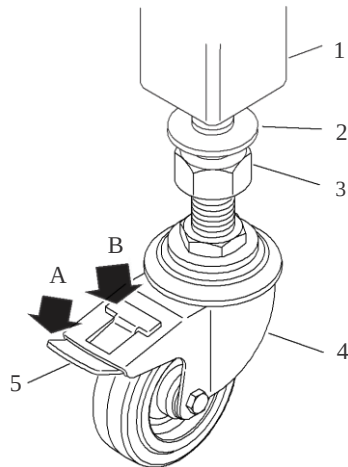
4.4.3 Locking and Unlocking the Casters



CAUTION

Do not use your hand or fingers to lock and unlock the casters. Use only protective footwear or an applicable tool to lock and unlock the lug for the caster wheel.

If your system has lockable casters (4) installed, you can apply the lock (5) to the caster wheel by using your foot or an applicable tool to push down on the lower lug (A). To release the lock on the caster wheel, push down the upper lug (B).



Item	Description
A	Lower lug (press to apply the lock)
B	Upper lug (press to release the lock)
1	Leg of the welded metal frame
2	Flat washer
3	Locknut
4	Adjustable height lockable caster
5	Locking lug for caster wheel

4.4.4 Adjusting the Height of the Lockable Casters



CAUTION

Do not use your hand or fingers to lock and unlock the casters. Use only protective footwear or an applicable tool to lock and unlock the lug for the caster wheel.

If your system is fitted with adjustable height and lockable casters (4), you can level the system and adjust the infeed and the outfeed heights of the conveyor by raising and lowering the casters (4).

Follow these steps to adjust the height of casters:

- 1 Make sure that all of the casters (4) are locked.
- 2 Use applicable lifting equipment to raise off the floor the two frame legs that have the casters you want to adjust.
- 3 Release the lock on the caster wheel you want to adjust by using an applicable tool to push down on the upper locking lug (B).
- 4 Slacken the locknut (3).
- 5 Use a spanner/wrench to rotate the adjustable height caster (4) in a clockwise direction, as necessary, to reduce the height of the frame (1).
- 6 Use a spanner/wrench to rotate the adjustable height caster (4) in a counterclockwise direction, as necessary, to increase the height of the frame (1).
- 7 Adjust the height of the other caster while you have the two frame legs raised.
- 8 When you have achieved the correct height for the adjustable height casters (4), turn the locknuts (3) in a clockwise direction until they attach the casters (4) tightly in the frame (1).
- 9 Lower the welded metal frame to the floor and check that the adjustment of the casters is correct.
- 10 Apply the lock to the caster wheels by using your shoe or an appropriate tool to push down on the lower locking lugs (A).
- 11 Adjust all of the adjustable height casters (4) until the welded metal frame is positioned firmly on the ground
- 12 Make sure that the adjustable height casters (4) evenly support the welded metal frame and the metal detection system so that it does not rock when the system operates.

4.5 Metal-Free Zone (MFZ)

To achieve optimum metal detector performance, make sure that an area surrounding the aperture of the metal detector, known as the Metal-Free Zone (MFZ), does not contain any type of metal.

The metal detector uses patented technology from METTLER TOLEDO to minimize the MFZ. This lets you to install the system in restricted spaces.

The size of the MFZ depends on the following criteria:

- Height of the aperture
- Type of metal detector
- Operating sensitivity of the metal detector
- Operating phase

Please consult the manual supplied with the metal detector installed on your system for information about the MFZ applicable to your metal detector.

4.6 Avoiding Mechanical Vibration



NOTICE

Do not attach the metal detection system to any other item of machinery as this could transmit vibrations to the metal detector and result in spurious signal triggering.

Follow these points to avoid mechanical vibration when you install the metal detection system:

- Do not support the system on or near vibrating structures or near vibrating machinery.
- Do not submit the system to any excessive vibration.
- Make sure that the frame construction of the metal detection system is stable and supported correctly.
- Keep all of the fasteners, nuts and bolts tight.
- Do not use the metal detector or the system as a convenient point to anchor any rigid or flexible conduit or pipework.
- Do not use rigid conduit of any type for the electrical power supply cable, unless required for the environment.

4.7 Connecting the Electrical Power Supply



DANGER

The safety circuit diagram supplied with the document pack is an outcome of a comprehensive risk assessment and qualification of the circuit components and architecture. The safety circuit should not be modified in any manner at any time without the express written consent of METTLER TOLEDO.

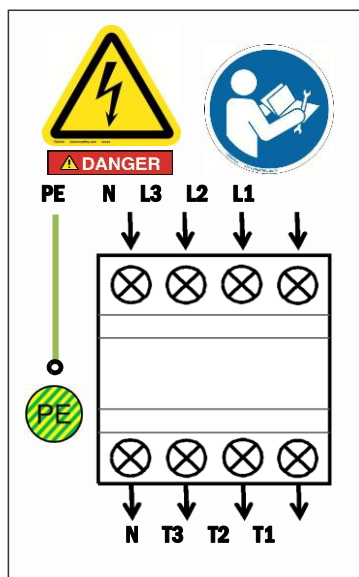
Modification of the safety circuit structure or components may cause death or injury to persons or damage to property.



DANGER

Only a qualified electrician or an electrical engineer is permitted to connect the system to the electrical power supply.

Connect the electrical supply by following the label installed on the system (see example below) and the electrical connection diagrams supplied in the document pack for the system.



Make sure that the electrical power supply is separated from other wiring. The earth/ground must be robust, of low impedance, noise free and assembled as shown in the figure in the 'Protected Earth/Ground' section.

Make sure that any dedicated supply line runs in its own conduit or trunking away from other noisy electrical power supplies.

Connecting the System to the Electrical Power Supply

The metal detection system is installed with a power supply isolator and a 24 VDC control circuit. The nameplate installed on the Industrial control enclosure of the system shows the power supply voltage for the system.



NOTICE

Do not use any type of rigid conduit for the electrical power supply cable, unless required.

Rigid conduit can transmit vibration to the metal detector and can cause spurious triggering.

There is a dedicated gland to install the electrical power supply cable to the system.

NOTE: The gland is located as close to the isolator as possible to make sure that the length of the supply cable is as short as possible.

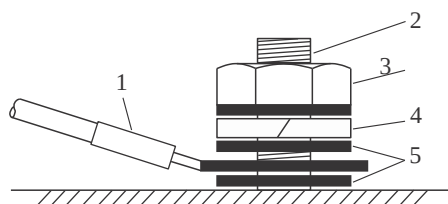
Nominal Voltages

- 1-phase, 85 VAC – 132 VAC, 48-63 Hz
 - **NOTE:** We recommend a 15 Amp D-rated MCB.
- 1-phase, 208 VAC – 230 VAC, 48-63 Hz
 - **NOTE:** We recommend a 10 Amp D-rated MCB.
- 3-phase, 342 VAC - 500 VAC, 50-60 Hz
 - **NOTE:** We recommend a 6 Amp D-rated MCB.

The Full Load Amperage (FLA) varies depending on the size of the motor.

Protective Earth/Ground

Connect the main protective earth/ground to the isolator or to a dedicated earth/ground stud. Follow the figure below for the protective earth/ground conductor termination.



Item	Description
1	Ring crimp
2	Earth/ground stud
3	Locknut
4	Spring washer
5	Flat washer

4.8 Connecting the Compressed Air Supply

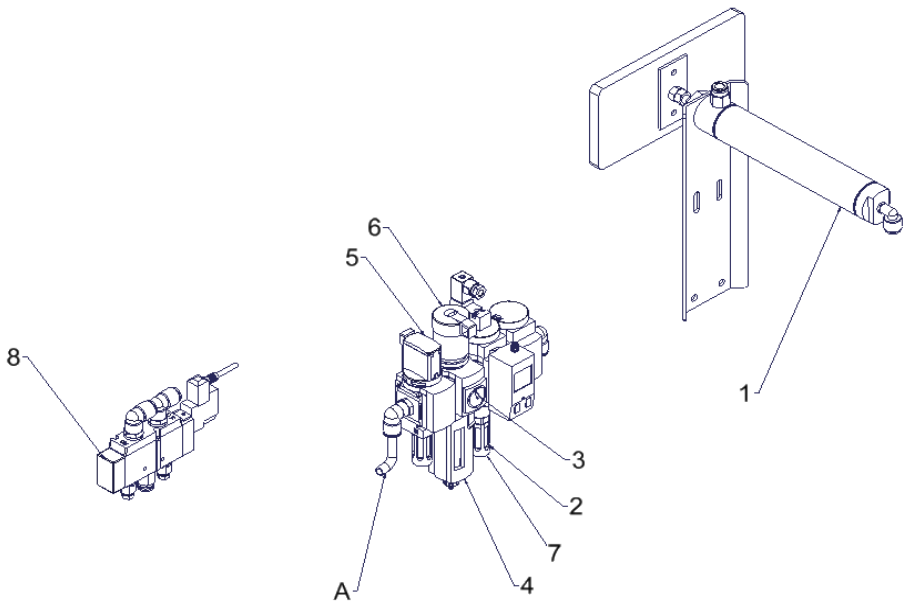
It is the responsibility of the installer to provide a supply of clean and dry compressed air at a pressure of (80 psi) if a pneumatic reject system is installed.

Make sure that the system's air supply is not shared with any other equipment.

Use the correct diameter of coupling for fitting the air supply hose to the connector on the system.

If someone presses the emergency stop button, or if someone opens a monitored access hatch or reject bin, the system automatically stops the conveyor belt, activates the electrically-operated pneumatic dump valve (2) and expels all of the air out of the automatic reject mechanism (1) to make it safe.

The following figure shows a typical arrangement of the components of a pneumatically-operated reject mechanism with a pusher reject device (1):



Item	Description	Item	Description
A	Air supply to the pneumatic system	5	Manual pneumatic shutoff/dump valve
1	Automatic reject mechanism	6	Air pressure regulator valve
2	Electrically-operated pneumatic dump valve	7	Air pressure switch
3	Air pressure gauge	8	Solenoid valve
4	Filter bowl		

To set the air pressure at (80 psi), lift the knob of the air pressure regulator valve (6), turn the knob in a clockwise or counterclockwise direction until the air pressure gauge (3) shows the correct air pressure, then press the knob down to lock it.

You can turn the manual pneumatic shutoff/dump valve (5) to stop the air supply (A) to the components of the pneumatic-operated reject mechanism (1) and to exhaust the air out of the system. The (Exhaust) text on the valve (5) shows that the compressed air supply is off. The (Supply) text on the valve (5) shows that the compressed air supply is on. The air pressure switch (7) checks that there is a compressed air supply to the solenoid valve (8).

4.9 Commissioning the System

When the metal detection system has been installed, METTLER TOLEDO can provide a commissioning service before you put the system into production.

5 Operation

This chapter explains how to:

- check the system before you start it;
- start the system;
- stop the system;
- use the emergency stop button;
- isolate the system's electrical power supply;
- isolate the compressed air supply to the automatic reject mechanism;
- understand warning systems and devices; and
- hand feed products onto the conveyor.

5.1 Before You Start the System

Do the tasks we describe in this section before you start the system or before every shift:

- Inspect the conveyor belt
- Check the guarding and all fasteners or hardware
- Examine the safety labels
- Inspect the electrical power supply cabling
- Check the sensors, light beams and cabling
- Inspect the liner of the metal detector's aperture
- Check the motor with gearbox unit
- Inspect the pneumatic system (automatic reject and due diligence systems only)



WARNING

Before you start the system, you must complete all of the safety-related tasks we describe in this section to avoid any personal injury or damage to the equipment.

5.1.1 Inspecting Flat PU and Modular Conveyor Belts and Rollers

NOTE: The best time to inspect the conveyor belt is after you have cleaned the system.

You need to inspect the condition of the conveyor belt when the belt is **not** running, as you cannot inspect the belt properly when the conveyor belt is moving too fast.

Follow these steps to inspect the conveyor belt and rollers:

- 1 Stop the system.
- 2 Remove all of the products from the conveyor belt.
- 3 Inspect the visible areas of the belt for signs of damage and dirt build-up.
- 4 For flat PU belts, inspect the joint for signs of damage and contamination.
- 5 Start and run the system until the rest of the belt that you were not able to inspect is visible.
- 6 Stop the system.
- 7 Inspect the rest of the conveyor belt.

If your system has a flat PU belt installed, inspect the edges of the belt for signs of fraying and examine the surface of the belt for:

- Discoloration from products;
- Signs of cracking (as particular cleaning solutions can cause the belt's surface to become brittle); and
- Signs of abrasion.



NOTICE

Inspect the flat PU belt and the infeed and drive rollers for any build-up of product.

If you allow a build-up of product on the rollers, the flat PU belt can overstretch, the rollers can collapse, and the belt can run off-center on the conveyor and cause damage to the metal detector's aperture and damage the edge of the belt.

If your system has a modular plastic conveyor belt, inspect the individual modules for signs of cracking, breakages and a build-up of dirt in-between the plastic linkages. You also need to make sure that the belt has not stretched due to wear on the interlocking pins.

5.1.2 Checking the Guarding and Fittings



WARNING

Never operate the system without the protective guards fitted.

Before you operate the system, do a visual check to make sure that the guards described below are not broken or cracked and that they are all in the correct position:

- Protective guarding for the infeed roller and outfeed roller
- Conveyor tunnel guards
- Automatic reject mechanism guarding

Do a visual check to make sure that all of the hardware on the system are in the correct locations.

5.1.3 Checking the Reject Bin

METTLER TOLEDO standard practice when providing a reject bin for retracting band conveyor systems, is to prevent the retracting band from actuating if the reject bin is out of place. The reject bin does prevent access to hazards when in the closed position. The bin is interlocked to detect its position, and access requires a tool to open the bin. When the bin is in the open condition, the conveyor will continue to run and the signal is removed to prevent retraction. If a reject event occurs, the conveyor will stop to prevent potential contaminant material from going downstream.

If the customer requires a retracting band conveyor system without a METTLER TOLEDO provided reject bin assembly, it is the customer's responsibility to apply safe integration practices, considering the residual risks from the METTLER TOLEDO system, the risks introduced by the customer's equipment, and the operational use of the equipment by the customer's employees. METTLER TOLEDO will identify the residual risks and safeguards applied, as well as interfaces to the METTLER TOLEDO safety circuit for the customer's safety inputs and outputs.

5.1.4 Examining the Safety Labels

Make sure that all of the safety labels and signal word labels are in the correct positions on the equipment and that the labels are clean, legible and not damaged.

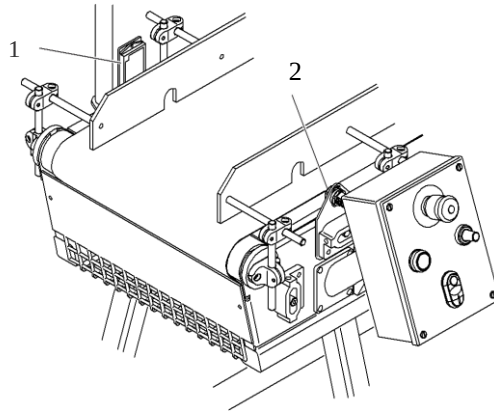
NOTE: A drawing that shows the types of safety labels and the correct positions of the labels is supplied with the General Assembly (GA) drawing of your system.

5.1.5 Checking the Electrical Power Supply Cabling

Make sure that the electrical power supply cable to the system is in good condition. Check for burns, wear and cuts to the cable shielding and also check for exposed and frayed wires and any other damage to the cable.

5.1.6 Checking Sensors, Light Beams and Cabling

Do a visual check to make sure that the sensors (1) with light beams (2) are not cracked or damaged and that the cabling is intact and not frayed.



5.1.7 Inspecting the Liner of the Metal Detector's Aperture

Inspect the condition of the liner of the metal detector's aperture for any damage caused by products or the conveyor belt. Also inspect the surface of the liner for any contamination and staining caused by ink from product packaging.

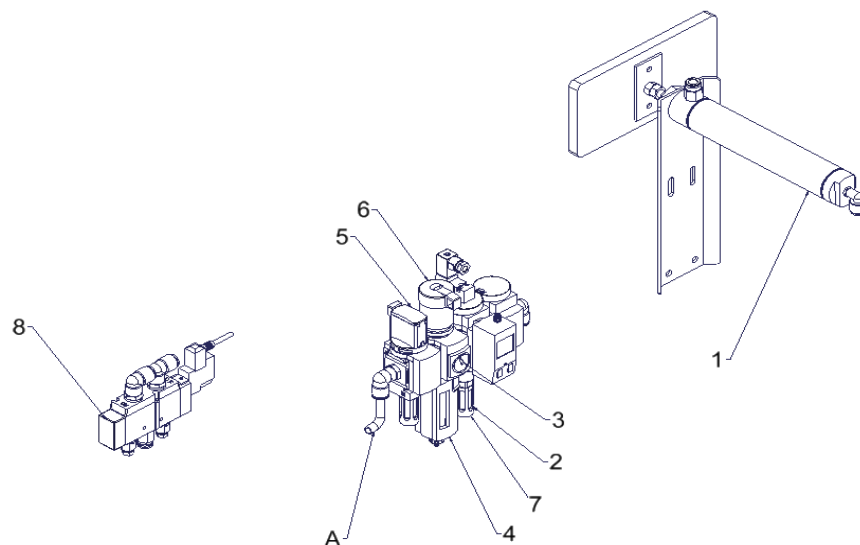
NOTE: This type of damage is normally caused when the conveyor belt is not aligned correctly on the conveyor or when the belt tracking is faulty.

5.1.8 Checking the Motor with Gearbox Unit

Every week, follow these steps to do a visual and manual inspection of the motor with gearbox unit:

- 1 Make sure that the vent for the rotating cooling fan is not blocked and that the fan rotates when the system is operating (if applicable.)
 - 2 Stop the system.
 - 3 Wait for approximately one hour for the motor with gearbox unit to cool down.
 - 4 Do a visual inspection of the entire unit. Look for obvious oil leaks on the floor of the production area below the motor.
 - 5 Put the palms of your hands all over the motor and gearbox unit and see if you can feel any oil leakage.
- ⇒ If you identify any oil leaks on the surface of the motor with gearbox unit or below the location of the motor, report this to service personnel and do not operate the system.

5.1.9 Inspecting the Pneumatic System



NOTE: A pneumatic system is installed only on a metal detection system that has an automatic reject mechanism.

Do a visual inspection all of the components of the pneumatic system to make sure that all of the hoses are located correctly.

Listen for air leaks and hissing sounds when the system is not in operation. Operate the system without products and listen for any obvious air leaks and hissing sounds from the pneumatic system.

Make sure that there is no build-up of liquid in the filter bowl (4). If the liquid is at a high level, ask service personnel to look at the pneumatic system and to empty the liquid out of the filter bowl (4).

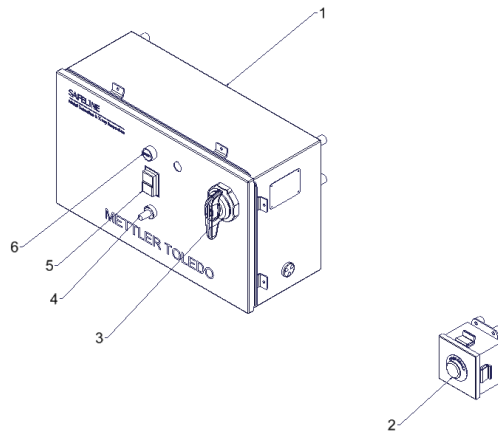
NOTE: The system may have an auto-drain that blows the liquid out of the filter bowl (4) onto the floor when it is full.

Check the air pressure gauge (3) and make sure that the air pressure in the system is (80 psi).

NOTE: If you have a heavy duty air blast reject mechanism, and you are using the system to inspect heavy products, the pressure needs to be at (80 psi).

NOTE: Refer to the specifications for your system for the correct operating pressure.

5.2 Starting, Stopping and Isolating the System



5.2.1 Starting and Stopping a Stop-on-Detect System

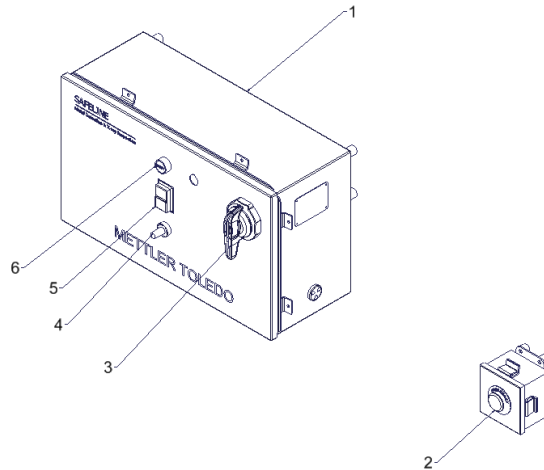
Item	Description
1	Metal detector power supply enclosure
2	Emergency stop button
3	Disconnect switch (on the control panel or the Industrial control enclosure)
4	Conveyor speed control (optional)
5	Start/stop conveyor buttons
6	Reset key switch or pushbutton (optional)

Follow these steps to start the conveyor on the stop-on-detect system:

- 1 Make sure that the disconnect switch (3) is on.
- 2 Make sure that the emergency stop button (2) is reset.
 - ⇒ If a warning beacon or an audible alarm is on, it means that the system has detected a contaminated product or there is a metal detector fault.
- 3 Remove any contaminated product(s) from the conveyor belt.
- 4 Turn the RESET key switch (6) or press the RESET pushbutton (6).
- 5 Press the start conveyor button (5) to start the conveyor.
- 6 To stop the conveyor, press the stop conveyor button (5).

NOTE: The system also stops when the metal detector identifies metal contamination or when someone presses the emergency stop button (2).

5.2.2 Starting and Stopping an Automatic Reject System



Follow these steps to start the conveyor on the automatic reject system:

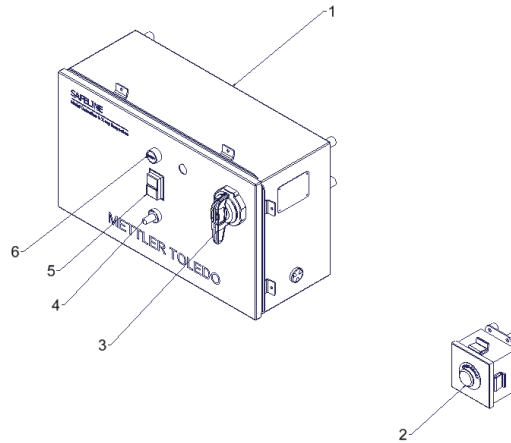
- 1 Make sure that the air supply is connected and working.
- 2 Make sure that the disconnect switch (3) is on.
- 3 Make sure that the emergency stop button (2) is reset.
- 4 If any of the warning beacons are on, reset the system. To do this, turn the Reset key switch (6) or press the Reset pushbutton (6).
- 5 Press the start conveyor button (5)
- 6 To stop the conveyor, press the stop conveyor button (5).

If your system has a safety relay, it causes the conveyor to stop when you open any tunnel guard.

The system also stops automatically when the following faults occur:

- Air pressure failure
- Metal detector fault
- Reject bin is full
- No confirmation of a rejected product
- Someone presses the emergency stop button
- If applicable, any safety-related relay is activated such as for an open access hatch

5.2.3 Starting and Stopping a Due Diligence System



Item	Description
1	Metal detector power supply enclosure
2	Emergency stop button
3	Disconnect switch (on the control panel or the Industrial control enclosure)
4	Conveyor speed control (optional)
5	Start/stop conveyor buttons
6	Reset key switch or pushbutton

Follow these steps to start the due diligence system:

- 1 Make sure that the air supply is connected and working
- 2 Make sure that the isolator (3) is on.
- 3 Make sure that the emergency stop button (2) is reset.
- 4 If any of the warning beacons are on, reset the system. To do this, turn the RESET key switch(6)or press the RESET pushbutton (6).
- 5 Press the start conveyor button (5)
- 6 To stop the conveyor, press the stop conveyor button(5).

If your system has a safety relay, it causes the conveyor to stop when you open any tunnel guard.

If someone opens the reject bin, the system exhausts the air out of the pneumatically-operated reject system but the conveyor belt continues. If any reject occurs while the reject bin is open, the Reject Confirmation Unit (RCU) stops the conveyor.

The system also stops automatically when the following faults occur:

- Air pressure failure
- Metal detector fault
- Reject bin is full
- No confirmation of a rejected product
- Someone presses the emergency stop button
- If applicable, any safety-related relay is activated such as for an open access hatch

5.2.4 Using the Emergency Stop Button



NOTICE

NEVER stop the system by pressing the emergency stop button unless there is an emergency situation.

Pressing the emergency stop button stops the conveyor and can also stop other systems on other production lines from operating.

The system has a large red emergency stop button installed on the control panel.

NOTE: Additional emergency stop buttons may also be installed on the frame.

If there is any kind of emergency that requires the metal detection system to stop immediately, push the emergency stop button.

The emergency stop button stays in a down position, the system stops operating, it automatically isolates the electrical power supply to the system and it activates the electrically-operated pneumatic dump valve to exhaust all of the air from the pneumatic system.

When the potential hazard has been resolved, you need to reset the emergency stop button. You can then press the start conveyor button to start the conveyor.

5.2.5 Isolating the System's Electrical Power Supply



DANGER

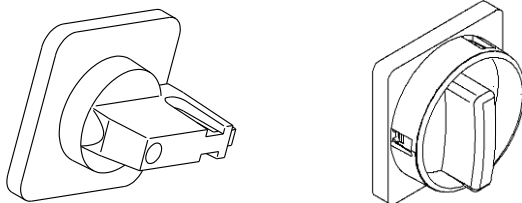
Do not open any electrical enclosure when the system is in operation.



DANGER

Only an electrician or an electrical engineer is permitted to open the main electrical enclosure. Be aware that the Variable Frequency Drive (VFD) stays live for up to five minutes after the electrical power supply has been disconnected.

NOTE: There are different types of disconnects (see examples in the figure below) but they all do the same task. The disconnect can be installed on the control panel or on the Industrial control enclosure.



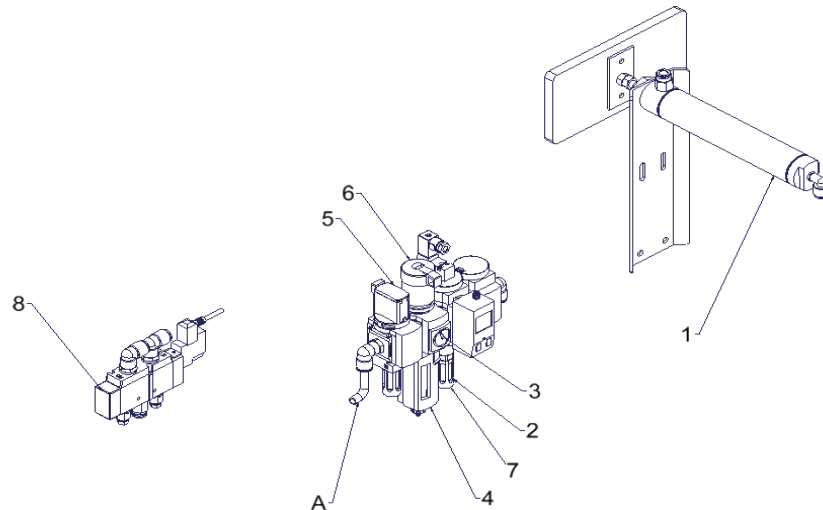
Follow these steps to disconnect the electrical power supply safely to the metal detection system for system maintenance and cleaning:

- 1 If the conveyor is running, press the stop conveyor button to stop the conveyor belt.
- 2 Turn off the electrical power switch.
- 3 If applicable, unplug the electrical power cord.
- 4 Turn the disconnect (power disconnect switch) in a counterclockwise direction to the OFF position.
- 5 If required, do the lockout and tagout procedure in accordance with your company's standard operating procedures.

Follow these steps to reconnect the electrical power supply to the system:

- 1 If applicable, plug in the electrical power cord.
- 2 Turn on the electrical power supply.
- 3 If required, do the lockout and tagout procedure in accordance with your company's standard operating procedures.
- 4 Turn the disconnect (power disconnect switch) in a clockwise direction to the ON position.
- 5 To start the conveyor, press the start conveyor button.

5.2.6 Isolating the Compressed Air Supply to the Reject Mechanism



Follow these steps to isolate the compressed air supply to the pneumatically-operated reject mechanism to make it safe for system maintenance and cleaning:

- 1 Turn the manual pneumatic shutoff/dump valve (5) to the (Exhaust) position.
⇒ The valve (5) exhausts all of the air out of the pneumatic system.
- 2 If required, do the lockout and tagout procedure in accordance with your company's standard operating procedures.

Follow these steps to reconnect the compressed air supply to the pneumatically-operated reject system:

- 1 If required, do the lockout and tagout procedure in accordance with your company's standard operating procedures.
- 2 Turn the manual pneumatic shutoff/dump valve (5) to the (Supply) position.
⇒ The air supply flows to the components of the pneumatically-operated reject system.

5.3 Understanding Warning Systems and Devices

This section explains the different types of faults and hazards that your system warns you about by using warning devices. It also explains the actions that you need to take to sort out particular faults and warnings.

5.3.1 Warning Systems for Stop-on-Detect Systems

A stop-on-detect system normally uses a single warning beacon. The beacon comes on and the conveyor stops when the metal detector identifies a contaminated product. If a SYSTEM RESET pushbutton is fitted on the control panel, operate it and then press the start conveyor button to restart the conveyor. If a SYSTEM RESET key switch is fitted, contact the authorized key holder.

5.3.2 Warning Systems for Automatic Reject and Due Diligence Systems

Up to five different colors of warning beacons can be joined together on top of each other in a stack to show you different types of faults and warnings as they occur when the automatic reject system or due diligence system is operating.



NOTE: The colors of the warning beacons in the table below are those that are normally used. Every beacon is normally marked with a label. However, the colors of the beacons on your system may have different meanings from these.

Warning beacon color	Meaning of the warning	Action needed
Amber (flashing)	Reject bin is full. The bin full sensor is blocked, or the lockable reject bin is not locked.	Contact the authorized key holder.
Blue (flashing)	Low air failure.	Contact system maintenance.
Red	Metal detector fault. For example, metal detector not in balance, sensor blocked or faulty or there is faulty cabling.	Contact system maintenance.

5.4 Hand-Feeding Products onto the Conveyor

This section is for systems when you need to manually feed products onto the conveyor instead of a slow-moving conveyor. Automatic reject systems and due diligence systems are normally fed automatically as they are part of the production line.

The way you feed the products onto the conveyor belt is based on how the metal detector is set up for a particular type of product. This is so that the detector can automatically 'learn' about a particular type of product.

Follow these main points when you hand-feed products onto the conveyor belt:

- Space the products equally on the conveyor belt.
- Be consistent in the way you position the products on the conveyor belt. For example, position them all with either the long or the short edge of the product facing the metal detector. Also, make sure that you position all of the products or packs the same way up.
- Position the products in the middle of the conveyor belt and make sure that they are not at an angle on the belt.
- Make sure that the products do not touch the liner of the metal detector's aperture.

If there is a large quantity of small products on the conveyor belt when the metal detector identifies a contaminated product and stops the belt, you need to remove all of products from the conveyor belt. Then you need to feed these products gradually through the metal detection system again until the system identifies only the contaminated products.

6 Routine Maintenance and Testing

This chapter explains all of the routine maintenance tasks that users need to do before they operate the system, or as and when necessary, during the operation of the system (depending on the type of products you are using the system to inspect).

It is important that you read and understand all of these routine maintenance procedures before you start the system and during the operation of the system, otherwise this can affect the condition of your equipment and affect the warranty.



DANGER

If you notice any faults with the system before or during operation, stop the system immediately and report any faults to the responsible person. Do not operate the system again until you have received authorization to start it. Not reporting faults to the responsible person can cause death, serious, moderate, or minor injury to a user or damage to the equipment.

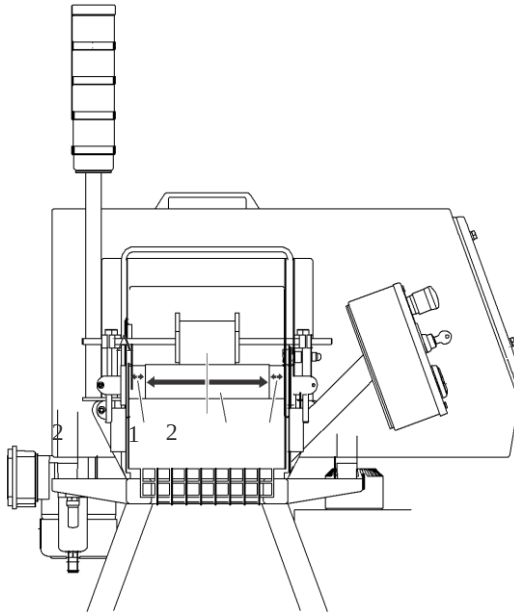
6.1 Testing the System During Production

Do the tasks we describe in this section at least once every day to test the system during production:

- **All systems**
 - Check the belt tracking
 - Test the operation of warning beacons and alarms
- **Automatic reject and due diligence systems**
 - Test the automatic reject mechanisms
 - Test the Reject Confirmation Unit (RCU) sensor
 - Test the bin full sensor

6.1.1 Checking the Belt Tracking

Operate the system and examine the horizontal position of the conveyor belt (1) during operation in relation to the skid plate (2). This is called the belt tracking. Make sure that the belt is running consistently at approximately the centerline of the conveyor (A).



NOTICE

Make sure that the belt runs at approximately the centerline of the conveyor.

If the belt runs into the surface of the liner in the metal detector, it will cause damage to the liner.

6.1.2 Testing Automatic Reject Mechanisms

To test the automatic reject mechanism installed on the system, you need to set the product parameters and run a test sample through the system with a product. Make sure that the reject mechanism operates and fully rejects the contaminated product from the production line into the reject bin. Make sure that the product enters the reject bin centrally.

6.1.3 Testing the Reject Confirmation Unit (RCU) sensor

The RCU sensor is positioned across the entrance of the reject bin. Quality Assurance personnel need to test the Reject Confirmation Unit (RCU) sensor before you run products through the system. You do this to make sure that the sensor works so that the system rejects products into the reject bin. If the RCU sensor is not working correctly, a contaminated product can get transported to the conveyor outfeed. Follow these steps to test the RCU sensor:

- 1 Stop the conveyor belt.
- 2 Make sure that the RCU sensor is on. To do this, check that the settings in the metal detector's screen show that the RCU sensor is fitted and calibrated.
- 3 Trigger the metal detector with a fault. To do this, block the RCU sensor and make sure that the system gives a message that there is an RCU fault.
- 4 If a warning beacon is installed, make sure that it flashes to show that there is an RCU fault.
- 5 Start the conveyor belt.
- 6 Put a contaminated test sample product across the infeed pack sensor to break the sensor beam and then through the metal detector's aperture. Do not allow the product to travel along the conveyor belt after the aperture. You do this to simulate a contaminated product traveling down the conveyor and not entering the reject bin.
- 7 Make sure that the system gives a message that there is an RCU fault.

6.2 Routine Performance Monitoring

This section explains how to routinely test the metal detection system using the applicable test samples. This procedure is called routine performance monitoring.

Who Tests the System?

This section is aimed at Quality Assurance (QA) staff.

Why Do I Need To Test the System?

The reason for testing is to make sure that:

- the metal detector continues to operate in accordance with the specified sensitivity standard;
- the complete metal detection system repeatedly and reliably detects and rejects contaminated products;
- all of the warning and signaling devices, such as alarm conditions and reject confirmation, are effective;
- you meet retailer codes of practice for auditing purposes;
- you are operating the system correctly; and
- you have the correct procedures in place.

You also need to test the system to make sure that changes do not occur when you are using the system such as:

- an air line is accidentally removed from a reject mechanism;
- the timer setting on the reject mechanism has been inadvertently altered;
- the metal detector's sensitivity has been turned down; and
- an operator called up the wrong product setting from the menu.

How Often Do I Need To Do a Test?

Your company should have standard operating procedures that dictate the frequency of testing. This is normally related to the ability to quarantine products inspected between performance monitoring routines. The frequency is normally short for perishable products and longer for non-perishable products.

How Do I Know When the Test Is Due?

Refer to the manual for your metal detector because, if routine performance monitoring is set correctly, the detector tells the user when the next test is due.

How Many Metal Types Do I Need To Test?

Metal is categorized into ferrous, non-ferrous and stainless steel for most applications. You need to check the metal detector's ability to detect all three types of metal.

Where Do I Put the Test Sample?

A metal detector's least sensitive point is the geometric center of the aperture. Therefore, you need to put the test sample in a position on the product so that it is as close as possible to the center of the aperture. Put the test sample in the center of the product. If this is not possible, put the test sample on top of the product.

Do not put the test sample on the conveyor belt with the pack or product on top of the sample as the sample will not pass through the area of least sensitivity. Passing the sample on the belt close to the coils of the metal detector also gives a false impression of the detector's true capability.

What Procedures Does the Test Need To Deal With?

At a minimum, the test needs to include the following:

- Test sample size, type and material used
- Position of the test sample in the aperture
- Effective use of test packs, if applicable
- Frequency of testing
- Number of tests
- Detector and reject devices test methods
- Due diligence systems testing
- Treatment of rejected and suspect products

What Records Do I Need To Keep Regarding Detector Testing?

You normally need to keep a record of the tests you do in accordance with your company's quality standards. These records also give the documented evidence that your customers and authoritative organizations such as Environmental Health Officers (EHOs) need. This is to show correct usage of the tests by following predetermined procedures.

You need to have documented procedures in place if a test fails with clear guidelines as to what action to take, by whom and when. Refer to the applicable retailer codes of practice for guidance.

6.3 Cleaning and Caring for the System

These instructions give personnel who clean and sanitize the system information about the metal detection system's materials and finishes.

The system user always has full and complete discretion, responsibility, and liability for determining the suitability of all cleaning and sanitization agents, protocols, and processes related to the system.

Regular cleaning is very important to keep good system operation. Proper cleaning and sanitization of equipment and work surfaces can:

- remove dirt and/or food material that harbor microorganisms;
- help reduce bacteria, including pathogens;
- help reduce risks of cross-contamination; and
- extend the operational life of the system.

Therefore, you need to do cleaning and remove product spillage in accordance with your company's appropriately-defined cleaning protocols. Only **trained and competent personnel** are permitted to sanitize the metal detection system.



WARNING

Cleaning metal detection machinery is permitted only when it is **OUT OF OPERATION** (stopped). The cleaning process can involve opening hinged lids and removing cover guards, if installed. Therefore, you must stop the system. Power off the system, unplug the power cord and disable all of the automatic reject mechanisms. Do lockout and tagout procedures in accordance with your company's standard operating procedures.



WARNING

Before you clean the system, wear sufficient protective clothing and hand protection as recommended by the manufacturer of any detergents used. In addition, **NEVER** wear loose clothing, watches, or jewelry during wash-down.

Standard metal detection systems are designed to meet the IP65 standard. This means that the systems will NOT withstand mid-pressure to high-pressure hose-down, but they WILL tolerate low-pressure (according to the IP65 standard) wash-down. Specifications for IP65 and optional standards related to water ingress are as follows:

- **IP65 (Standard Supplied System)** – Protected against water jets at a flow of 3.3 gal/min (12.5 L/min) and a pressure of 0.3 bar (4.35 psi or 30 kN/m²) – LOW PRESSURE.
- **IP66 (Standard Supplied System)** – Protected against temporary flooding of water and powerful jets of water (limited ingress permitted) of 26.4 gal/min (100 L/min) and 100kPa (pressure).
- **IP69 (Optional for Harsh Environments)** – Protected against water jets at a flow of 3.69–4.22 gal/min (14–16 L/min) and a pressure of 80–100 bar (1160–1450 psi or 8000–10,000 kN/m²) and a temperature of 80°C (176°F).

As well as the high-pressure water or steam wash described above, metal detection systems designed to the optional IP69K standard can withstand a dry steam wash with a maximum temperature of 180°C (356°F) and a pressure of 8 bar (116 psi or 800 kN/m²).



NOTICE

Use minimal amounts of water and pressure at all times.

6.3.1 Stopping the System and Doing Lockout and Tagout

Before you start any cleaning, stop the system and the conveyor and do lockout and tagout to isolate the system from the air supply and from the electrical power supply. Please refer to sections 'Isolating the Air Supply to the Reject Mechanism' and 'Isolating the System's Electrical Power Supply' for information about how to do this.

6.3.2 Using Detergents

Some components of the system can be susceptible to attack by some detergents or disinfectants. This can be detrimental to the operation of the system and can also shorten the effective life of the system.

Take care when you select detergents and disinfectants for use on various surfaces of the system (see information later in this section). On the basis of pH value, detergents can be classified as follows:

- **Acid detergents** – Detergents with a pH value below 4 are potentially damaging to the equipment. They are not suitable for daily cleaning so restrict their use to once a month at most, followed by a thorough rinsing process with clean water.
- **Neutral detergents** – Use preferably neutral detergents with a pH value of 7.
- **Alkaline detergents** – Alkaline detergents with a pH value between 7 and 12 are normally applicable to everyday cleaning.



CAUTION

If any of the above detergents are mixed with chlorine or a strong oxidizing agent (used to break down proteins and some vegetable staining such as carrot), then take special care to use the correct concentration and to thoroughly rinse it off

Clean system before first start up. Typical concentrations must not exceed 2%. Rinse off the concentrations within the specified time, typically 10 to 15 minutes and not more than 20 to 30 minutes. Never let the concentrations dry on the surface. Most detergents are sprayed on as a gel or foam, so it is important to thoroughly rinse the detergent off. This also applies to the disinfectant because, if it is left to dry on a surface, it increases the effect of the oxidizing agent in the detergent.

Machine surfaces

- **Stainless steel** - Use mild, non-abrasive acidic and alkaline cleaners. Do not use hydrochloric acid or chlorides.
- **Plastic** - More corrosion-resistant than stainless steel, plastics are resistant to chlorine but can crack or cloud from prolonged exposure to strong acidic or alkaline cleaners, and they are easily scratched. Use mild, non-abrasive acidic and alkaline cleaners.

6.3.3 Cleaning Procedure

You need to do the cleaning procedure consistently and in accordance with your company's appropriately- defined cleaning protocols. The user has full and complete discretion, responsibility, and liability to establish and do its cleaning protocols. The user can develop cleaning protocols based on the specific application for which the system is used.

6.3.3.1 Recommended Precautions

Follow these precautions before you start the wash-down procedure:

- Appropriately cover electrical equipment and other surrounding equipment that can get damaged by the cleaning process.
- Cover the motor with gearbox unit and the air intake parts with plastic.
- Unplug the system from the electrical power supply.
- Remove the conveyor belt (system maintenance personnel need to do this).

6.3.3.2 Wash-down Procedure

Follow this advice before you start any wash-down procedure:

- Always keep the protective cover installed over the screen of the metal detector.
- Replace any broken, worn, or missing reject mechanism and tunnel guarding on the system.
- Check the RTV 162 silicone adhesive sealant around the bezel of the heavy-duty liner. During wash- down, do not direct high-pressure water directly on the bezel of the liner. The silicone adhesive sealant needs to fill all gaps between inset liner surfaces. Replace the silicone adhesive sealant, as needed.
- Inspect the polypropylene flange and liner within the detector aperture for any apparent damage or deterioration.
- Inspect the rubber gasket between the detector access plate and detector on all remote metal detectors. The wash-down procedure is broken down into the following four main steps.

NOTE: Please consult the manufacturer's instructions for detergent use and pH level, and for any recommended protective clothing.

1. **Remove product residue and coarse dirt** – Remove any food items and other materials from the area you want to clean. If a reject bin is installed, ask an authorized key holder to remove any rejected products. Open the hinged lid on the side of the reject bin and then remove all product spillage and coarse dirt from on and around the reject area. You can use low air pressure to remove loose dirt or product from on and under the transport conveyors. **NOTE:** If an electronic interlock is installed on the reject cover it will dump the air from any pneumatic reject devices.

2. **Apply detergent** – Apply the detergent either manually or under low pressure. Use brushes for manual application to help you to remove stubborn or "invisible" dirt. This reduces the potential for cross-contamination and prevents detergent from drying on equipment surfaces. **DO NOT** use any abrasive tools that can damage the system's components.
3. **Allow time for detergent to become effective** – The time required for the detergent to become effective and the concentration depends on the detergent you use. Always obey the manufacturer's instructions and recommendations. As soon as the time has elapsed, start the final wash-down (the next main step). If activated detergent is left on for longer than the recommended time, the concentration will increase as the water evaporates and can cause discoloration of any stainless steel surfaces. With some activated detergents, this effect is greater and can vary depending on the type of stainless steel. Also, the rougher the surface, the greater can be the effect of the discoloration, because the detergent will stay on that surface longer. The rough surface can be caused by product particles adhering to it, or even sprayed-on disinfectant that is not rinsed off and allowed to dry.
4. **Wash down and final rinsing** – The wash-down and final rinse are very important to prevent the discoloration or damage to metal surfaces. Although there is a temptation in wash-down environments to use high-pressure water jets, we stress that only low pressure is allowed. Start from the top of the system and work down and in the direction of the product flow.

Always inspect the safety labels after wash-down. Replace any worn or missing labels.

NOTE: You can order any worn or missing the labels by using the part numbers shown in the document pack supplied with the system.

The user is responsible to establish cleaning protocols for their individual systems. Users can decide that such cleaning protocols require that:

- someone who is not involved in the actual cleaning process does the cleaning inspection; and
- this person signs and dates an established machine cleaning register to show that the cleaning has been done correctly.

6.3.4 Restarting the System after Sanitization

When you have finished the sanitization process, remove all lockout and tagout devices and return the system to its normal operation condition. Before you restart the system, make sure that all safety devices are back in their original positions and correctly fastened. For example, if you removed any guards for cleaning, make sure that you reinstall them.



WARNING

NEVER start the system without all original safety features in the correct locations.

Re-engage all interlocks before you start the system. On some systems, a blue fault lamp in the SYSTEM RESET button comes on until all interlocks are engaged.

After cleaning, check all cables connectors and compressed air hoses for leakage, loose connections, rub marks and damage. Tighten, repair or replace any faulty cables and air hoses, as necessary.

Disclaimer: The end user shall at all times keep full and complete discretion, responsibility, and liability for determining the suitability of all cleaning and sanitization agents, protocols, and processes of whatsoever kind or type related to the equipment. At the end user's request, METTLER TOLEDO shall provide the end user with information about the materials and finishes which comprise the equipment, such as the grade of stainless steel.

6.4 Storing the System

If you do not operate the system for some time and want to store it until you use it again, you need to do the steps that follow:

- 1 Clean and sanitize the system.
- 2 Make sure that every part of the system is dry.
- 3 Wrap the system in cling film or a similar packaging material.
- 4 Add applicable moisture-absorbing packages, such as Dessicant (Silica Gel) inside the wrapped system.
Store the system in a dust-free and dry location in accordance with the environmental conditions below:
 - ⇒ Temperature range: -10 degrees Celcius to 40 degrees Celcius
 - ⇒ Relative humidity: 80% up to 31 degrees Celcius (86 degrees Farenheit) reducing to 50% at 40 degrees Celcius (104 degrees Farenheit)

If you want to store the system while it is still packed in the crate after you have received it, follow step 5 above.

6.5 Disposal

In conformance with the European Directive 2012/19/EU on Waste Electrical and Electronic Equipment (WEEE) this device may not be disposed of in domestic waste. This also applies to countries outside the EU, per their specific requirements.



Please dispose of this product in accordance with local regulations at the collecting point specified for electrical and electronic equipment. If you have any questions, please contact the responsible authority or the distributor from which you purchased this device. Should this device be passed on to other parties, the content of this regulation must also be related.

7 Troubleshooting

This chapter gives users a short guide to resolving common and minor faults with the system. We advise you to follow the instructions related to particular faults we describe before you contact maintenance personnel.

7.1 Conveyor Does Not Run

Follow these steps to sort out minor faults if the conveyor does not run. Press the start conveyor button after each step and, if the conveyor does not start, move to the next step.

- 1 Make sure that there is power to the system. To do this, check that the screen is on and that the system is plugged in at the electrical power supply.
- 2 Turn the disconnect switch to the ON position.
- 3 Reset the red emergency stop button.
- 4 Make sure that the sensors are not blocked by any products or any other objects.
- 5 Make sure that the tunnel guards are in the correct positions.
- 6 If an electro-mechanical lock is installed on the reject bin, make sure that the reject bin door is properly closed.
- 7 If a RESET pushbutton or key switch is installed on the control panel, press the SYSTEM RESET Push button or ask an authorized key holder to turn the SYSTEM RESET key switch.
- 8 If a blue warning beacon is installed, check if it is flashing. If it is flashing, it means that the compressed air supply to the system's automatic reject mechanism is too low.
- 9 Check the air pressure gauge. If it gives a zero indication, it means that there is no compressed air supply to the system. To repair this fault, turn on the manual pneumatic shutoff/dump valve to the SUP (Supply) position to open the air supply to the system.
- 10 Check the air pressure gauge again. If it still gives a zero indication, turn on the compressed air supply to the system.
- 11 Make sure that the air pressure gauge gives an indication (80 psi).
- 12 Press the start conveyor button. If the conveyor still does not start, contact system maintenance personnel.

7.2 Testing the Conveyor Belt for Metal Contamination

Sometimes, the conveyor belt may get contaminated and repeatedly give a false reject signal on the metal detector's screen. If this happens, notify your supervisor so they can ask Service or Quality Assurance to do the test that follows:

- 1 Stop the conveyor belt.
- 2 Make sure that the repeated false reject signal on the screen stops. You do this to make sure that the metal detector is not faulty.
- 3 Start the conveyor belt.
- 4 When the screen shows a false reject, put a removable pen mark on the surface of the belt immediately with your hand positioned as close as possible to the aperture at the conveyor infeed.
- 5 Repeat the action in the step above about eight times.
- 6 Make sure that the marks you make on the surface of the belt are in approximately the same horizontal position across the belt.
- 7 Depending on the speed of the conveyor belt, make an estimated calculation of the position of the contamination on the belt. This is the approximate distance that the belt travelled through the aperture after the screen showed a false reject and you made a mark on the belt.
- 8 Remove and inspect the conveyor belt for contamination in the area when you may know the location.

8 Technical Data

Description	Specification
Dimensions	Each system is designed to match a customer's requirements.
Power requirements	
Standard system option	1-phase, 85 VAC – 132 VAC, 48-63 Hz
	NOTE: We recommend a 15 Amp D-rated MCB
	1-phase, 208 VAC – 230 VAC, 48-63 Hz
	NOTE: We recommend a 6 Amp D-rated MCB.
	3-phase, 342 VAC - 500 VAC, 50-60 Hz
	NOTE: We recommend a 6 Amp D-rated MCB.
Control voltage	24 VDC
Current consumption	Varied depending on the size of the motor.
Air supply (pusher and retracting reject mechanisms)	
Maximum pressure	80 psi
Capacity (liter/second at 80 psi)	Varies depending on the type of automatic reject system installed.
Environmental conditions	
Operating temperature range	-10 degrees Celcius to 40 degrees Celcius.
Relative humidity	80% up to 31 degrees Celcius (86 degrees Farenheit) reducing to 50% at 40 degrees Celcius (104 degrees Farenheit).

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